Consolidated English research manuscript

Arithmetic Transcendence of Integrals, Quantiles, and Lemniscates

E-functions, cyclotomic periods, local logarithms,
and the obstruction at Euler's constant

KEIO

11 October 2026

Abstract. We assemble a sequence of investigations into the arithmetic nature of explicit integrals, inverse distribution functions, and algebraic-curve lengths. An antisymmetric iterated exponential integral is treated by differential-field translations and *E*-function specialization. Reparametrized Gaussian integrals give a finite classification of algebraic quantiles of truncated mixtures. Logarithmic forms yield arithmetic classifications for Student distributions, half-integral beta distributions, and shortest beta probability intervals. A square-root cover turns polynomial lemniscate lengths into closed periods; a cyclotomic homology argument treats rational arc divisions without requiring simple Jacobians. Algebraic division ratios are reduced to torsion, and local abelian logarithms combined with primitive divisors give rational-input and bounded-degree results. Fermat-quotient torsion classifications, including a generalized-Jacobian treatment of second-kind differentials, provide stability under polynomial weights for a family of incomplete beta integrals. Finally, we examine the attempted transfer to the Euler–Mascheroni constant. Explicit coefficient locking, an essential-singularity obstruction, disjoint local convergence domains, and a missing arithmetic zero-division principle explain why the preceding results do not constitute a proof for that constant.

Status. This is a unified proof manuscript, not a claim of established publication priority or completed independent peer review. Classical inputs are cited separately from the arguments assembled here. The paper does *not* prove the irrationality or transcendence of the Euler–Mascheroni constant. Numerical certificates check computations, not the external transcendence theorems or the logical correctness of every proof.

Keywords. *E*-function; analytic subgroup theorem; generalized Jacobian; incomplete beta function; quantile; lemniscate; primitive divisor; Euler–Mascheroni constant.

Single-file LaTeX edition. All formulas, tables, references, and the
representative verification listing are contained in this source.

Contents

1 Introduction and logical organization **3**
1.1 Principal families and their scope . . . 3
1.2 Conventions and distinctions . . . 3
1.3 Proven inputs, manuscript claims, and conditional conclusions . . . 4

2 External inputs and a reusable separation principle **4**
2.1 Arithmetic differential functions . . . 4
2.2 Logarithms and algebraic groups . . . 4
2.3 Other inputs used later . . . 5

3 Antisymmetric iterated exponential integrals **5**
3.1 A distinguished constant and an infinite base field . . . 7
3.2 An exact, but insufficient, connection to Euler's constant . . . 7

4 Gaussian endpoint separation and finite quantile classification **7**
4.1 The endpoint-vector algorithm . . . 8
4.2 Two explicit transcendental medians and genuine exceptions . . . 9
4.3 Why this does not settle a specified standard normal quantile . . . 9

5 Logarithmic forms and statistical quantiles **9**
5.1 A parity classification for Student distributions . . . 9
5.2 Half-integral beta laws and three possible algebraic pairs . . . 10
5.3 Binomial intervals and the F distribution . . . 12

6 Shortest beta probability intervals **12**
6.1 Existence, uniqueness, and the density constraint . . . 12
6.2 A monotone correction divided by the density . . . 13
6.3 Transcendence of endpoints, width, and midpoint . . . 13
6.4 Examples and posterior interpretation . . . 14

7 Polynomial lemniscates: lengths as closed periods **15**
7.1 Removing powers and constructing one connected curve . . . 15
7.2 Critical points are crossings, not cusps . . . 15
7.3 The positive square root closes on an algebraic cover . . . 16
7.4 Global holomorphy and the exact circle exception . . . 16
7.5 A necessary counterexample outside the hypothesis . . . 17
7.6 A convergent certificate for a concrete normalized length . . . 17

8 Cyclotomic homology and all rational arc divisions **17**
8.1 The primitive cyclotomic factor . . . 18
8.2 A second integral contradicts proportional division . . . 18
8.3 The small-exponent algebraic cases . . . 19
8.4 The prime-only route and why it was superseded . . . 19
8.5 Geometric division points and their boundary exceptions . . . 19

9 Algebraic division ratios and torsion **20**
9.1 The reference endpoint is always torsion . . . 21
9.2 Complete exclusion for $N = 8, p, 2p$ . . . 21
9.3 General algebraic exceptions: fields, conjugates, and finiteness . . . 22
9.4 Two algebraic irrational ratios for every exponent . . . 22

10 Local abelian logarithms and primitive divisors **22**
10.1 All rational locations, without an exponent exception . . . 23

11 Uniformity for bounded algebraic degree **24**
11.1 A small-exponent ramified quadratic example . . . 25

12 Fermat beta integrals, polynomial weights, and second-kind forms **25**

12.1 The curve and the pole that cannot be ignored . . . 26
12.2 Generalized Jacobians retain the polar part . . . 26
12.3 A nonzero period for a possibly second-kind form . . . 27
12.4 Projection rigidity, not vanishing of the entire logarithm . . . 27
12.5 Torsion packets force three-term independence . . . 28
12.6 Exact polynomial reduction and its algebraic exception . . . 28
12.7 The exact linear dimension of all weighted values . . . 29
12.8 Explicit first-kind and second-kind examples . . . 29

13 Attempted transfer to the Euler–Mascheroni constant 30
13.1 A normalization ambiguity invisible to beta quotients . . . 30
13.2 Confluence returns gamma, but not an arithmetic lower bound . . . 30
13.3 Polynomial logarithmic moments lock the gamma and tail coefficients . . . 31
13.4 Why a finite modulus does not directly accommodate the exponential . . . 32
13.5 Two arithmetic blocks containing the target exactly . . . 32
13.6 An arithmetic division obstruction and the missing implication . . . 33
13.7 The two direct local series have disjoint convergence domains . . . 34
13.8 A genuine conditional conclusion, kept conditional . . . 34
13.9 What survives the attempted transfer . . . 35

14 Numerical certification, reproducibility, and limitations 35
14.1 Positive series and explicit remainder estimates . . . 35
14.2 Verification scripts rerun for the unified edition . . . 36
14.3 Representative certification scales . . . 37

15 Synthesis and unresolved questions 37

A Catalogue of explicit constants 38

B Concordance with the research notes 39

C A compact exact-arithmetic verifier 40

References 43

1 Introduction and logical organization

The common problem in this paper is to identify an arithmetic relation that would have to exist if a specified real number were algebraic, and then to exclude that relation by a theorem whose hypotheses can actually be checked. The relevant relation is not the same in every example. For an E-function value it is a functional algebraic relation; for a beta quantile it may be a linear form in logarithms; for an abelian integral it can be a torsion condition; for an arc length it is a closed-period representation. Confusing these mechanisms is especially dangerous when attempting to pass to a limit.

The organization is thematic rather than chronological. Earlier special cases are retained where they explain a hypothesis, but later extensions are stated only once. In particular, the prime-leaf argument is subsumed by the cyclotomic argument for all integral exponents, and the first-kind beta argument is subsumed by its second-kind extension. An explicit concordance with the source research notes is provided in Section B.

1.1 Principal families and their scope

Family	Conclusion developed in this manuscript	Principal input
Antisymmetric Ein integrals	Simultaneous algebraic independence, including an additional iterated-integral coordinate	Differential algebra and Siegel–Shidlovskii
Truncated Gaussian mixtures	At most $5m^2$ candidate locations for all algebraic location–probability pairs	Beukers specialization
Student and half-integral beta laws	Parity classification and a three-pair exception criterion	Baker and Gelfond–Schneider
Shortest beta intervals	Integer/noninteger arithmetic dichotomy for rational parameters of integral sum	Logarithmic forms and a monotonicity lemma
Polynomial lemniscates	Transcendence of length; separation from logarithms unless the curve is a circle	Closed periods and algebraic groups
Cyclotomic arc divisions	Rational divisions classified for every exponent; algebraic ratios constrained by torsion	Analytic subgroup theorem
Local and bounded-degree inputs	Uniform rational-input results and an explicit height-independent large-exponent bound	Local logarithms and primitive divisors
Fermat beta integrals	Linear independence stable under arbitrary polynomial weights, with an exact algebraic exception	Torsion packets and generalized Jacobians
Euler's constant	Exact identities and obstructions, but no arithmetic classification of γ	Mixed arithmetic Gevrey framework

1.2 Conventions and distinctions

The field $\overline{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$. An algebraic real number belongs to $\overline{\mathbb{Q}} \cap \mathbb{R}$. The phrase "the quantile is transcendental" always refers to the exact inverse-function value, not to a terminating printed decimal. All real powers of positive numbers use their positive real branches; complex logarithms are fixed on the relevant path.

A period is an integral over an integral or rational homology class. An abelian logarithm is a vector in a complex Lie algebra, not merely its class modulo a period lattice. Whenever individual coordinates are compared, actual paths and actual logarithm vectors are fixed. Torsion means finite order in the specified algebraic group; torsion of a projected point is not silently identified with torsion in the original Jacobian.

The symbols γ and Γ have distinct meanings:

$$\gamma = \lim_{n\to\infty}(H_n - \log n) = -\Gamma'(1), \qquad H_n = \sum_{k=1}^{n}\frac{1}{k}, \quad H_0 = 0. \tag{1.1}$$

The differential transcendence of the function Γ does not settle the arithmetic of $\Gamma'(1)$. Also, linear independence, algebraic independence, and the transcendence of a single number are different conclusions. They are distinguished throughout.

1.3 Proven inputs, manuscript claims, and conditional conclusions

Theorems attributed to named authors are external inputs. The remaining numbered statements have proofs or explicit reductions below; their inclusion records the claims of this consolidated research draft, not an assertion of independent validation or novelty. No theorem below about quantiles or curves uses an unproved claim about γ. The only implications involving unproved arithmetic separation principles are marked as conditional in Section 13.

Tamiozzo's Remarks 5.7 and 8.16 concern, respectively, division points and lengths of polynomial lemniscates [7]. The proposed arguments in Sections 7 and 8 address those formulations. They should not be confused with Erdős's extremal-length problem. This manuscript does not establish a historical first solution, and the absence of an identical statement in a literature search is not evidence of priority.

2 External inputs and a reusable separation principle

2.1 Arithmetic differential functions

An E-function is an entire series $f(z) = \sum a_n z^n/n!$ with algebraic coefficients, exponentially bounded conjugates and common denominators, and a linear differential equation over $\overline{\mathbb{Q}}(z)$. In the applications below, these conditions are checked by coefficient formulas and a displayed finite first-order system.

External theorem 2.1 (Siegel–Shidlovskii and Beukers)**.** *Let a vector of E-functions satisfy $Y' = M(z)Y$ over $\overline{\mathbb{Q}}(z)$. At a nonzero algebraic point which is not a pole of M, the transcendence degree of the values equals that of the functions over $\overline{\mathbb{Q}}(z)$. Moreover, a homogeneous algebraic relation among the values lifts to a functional relation of the same degree. In particular, functional linear independence implies linear independence of the values at such a point.*

The version and the linear consequence used here are Theorems 1.1 and 1.3 and Corollary 1.4 of Beukers [1].

External theorem 2.2 (Kolchin–Ostrowski, antiderivative form)**.** *Let a differential field of characteristic zero have algebraically closed constant field C. In an extension with no new constants, suppose $f_1', \dots, f_m'$ belong to the base field. If the f_i are algebraically dependent over that field, some nonzero C-linear combination of them belongs to it.*

A convenient reference is Srinivasan [2]. The derivative hypothesis is imposed at a single level; it is not applied indiscriminately to iterated antiderivatives at different levels.

2.2 Logarithms and algebraic groups

We use Hermite–Lindemann in the form $0 \neq a \in \overline{\mathbb{Q}} \Rightarrow e^a \notin \overline{\mathbb{Q}}$, and Gelfond–Schneider in the form $a^b \notin \overline{\mathbb{Q}}$ for algebraic $a \neq 0, 1$ and algebraic irrational b, with any fixed logarithm of a. Baker's theorem

says that an algebraic-coefficient linear form in fixed logarithms of nonzero algebraic numbers is either zero or transcendental [3].

External theorem 2.3 (Analytic subgroup theorem)**.** *Let $G/\overline{\mathbb{Q}}$ be a connected commutative algebraic group, let $u \in \mathrm{Lie}(G)_{\mathbb{C}}$ satisfy $\exp_G(u) \in G(\overline{\mathbb{Q}})$, and let V be a $\overline{\mathbb{Q}}$-linear subspace with $u \in V_{\mathbb{C}}$. There is a connected algebraic subgroup $H \subseteq G$ such that*

$$u \in \mathrm{Lie}(H)_{\mathbb{C}} \subseteq V_{\mathbb{C}}.$$

The statement includes logarithms whose exponential is the identity.

We use the strengthened formulation in Huber–Wüstholz, Theorem 6.2 [4]. A related consequence is that a nonzero closed period of an algebraic rational differential on a smooth connected projective curve is transcendental; the cycle must avoid its poles. See [4, Theorem 13.9 and Corollary 13.13].

Lemma 2.4 (Abelian versus affine logarithms)**.** *Let $A/\overline{\mathbb{Q}}$ be an abelian variety, let u be a logarithm of an algebraic point, and let λ be an algebraic invariant differential. If $\lambda(u) \neq 0$, then*

$$\lambda(u) \notin \overline{\mathbb{Q}} + \mathrm{span}_{\overline{\mathbb{Q}}}\{z \in \mathbb{C} : e^z \in \overline{\mathbb{Q}}^{\times}\}. \tag{2.1}$$

Proof. Suppose $\lambda(u) = b_0 + \sum b_j z_j$ with algebraic b_j and algebraic e^{z_j}. Apply External theorem 2.3 to $A \times \mathbb{G}_a \times \mathbb{G}_m^r$, the logarithm $(u, 1, z_1, \dots, z_r)$, and the corresponding hyperplane. A connected algebraic subgroup of $A \times L$, where L is connected affine and commutative, is a product $B \times D$. Indeed, its connected affine part maps trivially to A; after dividing by that part, its abelian quotient maps trivially to the affine quotient of L. These are the standard consequences of the Barsotti–Chevalley theorem [5]. Consequently $(u, 0, \dots, 0)$ belongs to the same subgroup Lie algebra. The hyperplane then gives $\lambda(u) = 0$, a contradiction. □

In particular a nonzero algebraic linear form in coordinates of abelian logarithms cannot be algebraic. Products of abelian varieties are allowed in this assertion.

2.3 Other inputs used later

The geometric torsion packets used in Sections 9 and 12 are those of Boxall–Grant [10] and Arul [11]; their exact special cases are stated where applied. Manin–Mumford is used only for finiteness, not for an effective enumeration [12]. The local logarithm is a functorial homomorphism on a finite extension of $\mathbb{Q}_\ell$, kills torsion, and agrees on a sufficiently small good-reduction residue disk with termwise integration [13]. Bang–Zsigmondy and Voutier provide primitive divisors in Section 10; their exceptional cases and numerical thresholds are kept explicit.

3 Antisymmetric iterated exponential integrals

Define the entire function

$$\mathrm{Ein}(z) = \int_0^z \frac{1 - e^{-t}}{t}\,\mathrm{d}t = \sum_{n \geq 1} \frac{(-1)^{n+1} z^n}{n\, n!}. \tag{3.1}$$

For positive integers $i < j$, put

$$F_i(z) = \mathrm{Ein}(iz), \qquad H_{ij}(z) = \int_0^z \big(F_j(t)F_i'(t) - F_i(t)F_j'(t)\big)\,\mathrm{d}t. \tag{3.2}$$

The integrands are entire, so these primitives are path independent. The antisymmetry matters: the analogous sum integrates to F_iF_j and produces no additional coordinate.

Theorem 3.1. *For every $m \geq 2$ and every $0 \neq \alpha \in \overline{\mathbb{Q}}$, the numbers*

$$e^{-\alpha}, \quad F_i(\alpha)\ (1 \leq i \leq m), \quad H_{ij}(\alpha)\ (1 \leq i < j \leq m)$$

are algebraically independent over $\overline{\mathbb{Q}}$. Their number is $1 + m + \binom{m}{2}$.

Proof. Set $u = e^{-z}$ and $K = \mathbb{C}(z, u)$, with derivation $D = \partial_z - u\partial_u$. The exponential is transcendental over $\mathbb{C}(z)$, as follows by comparing the largest exponential along the negative real axis in a hypothetical polynomial relation. Thus K is the rational function field $\mathbb{C}(u)(z)$.

For $R \in K$, the derivative DR cannot have exactly a simple pole at the formal divisor $z = 0$. If $R = r_{-a}(u)z^{-a} + \cdots$, $a \geq 1$, then its derivative has leading term $-ar_{-a}(u)z^{-a-1}$. If R is regular there, so is DR. This is a valuation with u treated as an independent coordinate, not the analytic limit along $u = e^{-z}$.

Since $F_i' = (1 - u^i)/z$, External theorem 2.2 would turn algebraic dependence of the F_i over K into

$$DR = \frac{\sum_i c_i(1 - u^i)}{z}, \qquad (c_i) \neq 0.$$

The numerator is a nonzero polynomial. The simple pole contradicts the preceding observation. Hence the F_i are algebraically independent.

Let $L = K(F_1, \ldots, F_m)$. All H_{ij}' lie in L. If the H_{ij} were algebraically dependent over L, another application of External theorem 2.2 would give

$$S = \sum_{i<j} c_{ij}H_{ij} \in L$$

with not all c_{ij} zero. For each k, let σ_k translate F_k by 1 and fix the other generators. These are commuting differential automorphisms of L, and differentiation gives

$$\sigma_k S - S = \sum_{i<k} c_{ik}F_i - \sum_{j>k} c_{kj}F_j + d_k, \qquad d_k \in \mathbb{C}.$$

For $i < j$, applying the two translations in opposite orders yields

$$(\sigma_i\sigma_j - \sigma_j\sigma_i)S = 2c_{ij}.$$

Commutativity forces every c_{ij} to vanish. Therefore u, all F_i, and all H_{ij} are algebraically independent over $\mathbb{C}(z)$.

To specialize, write $H_{ij}(z) = \sum a_n^{ij} z^n/n!$. Direct convolution shows

$$a_n^{ij} = \frac{(-1)^n}{n} \sum_{k=1}^{n-1} \binom{n}{k} \frac{j^k i^{n-k} - i^k j^{n-k}}{k}. \tag{3.3}$$

If $L_n = \operatorname{lcm}(1, \ldots, n)$, then

$$|a_n^{ij}| \leq 2(i + j)^n, \qquad \operatorname{den}(a_n^{ij}) \mid L_n^2.$$

The standard estimate $\log L_n = O(n)$ gives the required exponential common-denominator bound. A finite differential system is obtained by including u^r for $0 \leq r \leq 2m$, the F_i, the products $u^k F_i$, and all H_{ij}; for example

$$(u^k F_i)' = -ku^k F_i + (u^k - u^{k+i})/z.$$

Its only finite pole is 0. All coordinates are E-functions and the auxiliary coordinates add no transcendence degree. Apply External theorem 2.1 at α. □

For one additional primitive, the implication "algebraic primitive implies it lies in the base field" can also be seen by taking the normalized trace: if $H' \in L$ and H is algebraic of degree d, then $R = d^{-1}\operatorname{Tr}(H)$ satisfies $R' = H'$. No new constants are introduced in the meromorphic function fields used above.

3.1 A distinguished constant and an infinite base field

Set

$$\mathcal{A} = H_{12}(1) = 0.06689807721166676186223537447755142651 99\ldots. \tag{3.4}$$

In particular $e, \mathrm{Ein}(1), \mathrm{Ein}(2), \mathcal{A}$ are algebraically independent. Since every polynomial involves only finitely many generators, Theorem 3.1 also gives

$$\mathcal{A} \text{ is transcendental over } \overline{\mathbb{Q}}(e, \mathrm{Ein}(1), \mathrm{Ein}(2), \mathrm{Ein}(3), \ldots). \tag{3.5}$$

For $F = F_1$, $G = F_2$, put $B = G - F$, $U = uG$, $V = u^2F$, and $H = H_{12}$. The eight functions $(1, u, u^2, u^3, B, U, V, H)$ form the system

$$(u^r)' = -ru^r \quad (1 \le r \le 3), \qquad B' = (u - u^2)/z, \tag{3.6}$$

$$U' = -U + (u - u^3)/z, \qquad V' = -2V + (u^2 - u^3)/z, \tag{3.7}$$

$$H' = (B - U + V)/z. \tag{3.8}$$

They are linearly independent over $\mathbb{Q}(z)$ by the already established algebraic independence. The classical quantitative linear-independence estimate recalled in [6], with dimension eight, gives

$$2 \le \mu(\mathcal{A}) \le 8, \tag{3.9}$$

where μ is the irrationality exponent. This is an upper bound, not an assertion that the exponent equals eight; it also excludes Liouville behavior.

3.2 An exact, but insufficient, connection to Euler's constant

For $r, R > 0$ define $H_r(R)$ as in (3.2), with scales $1, r$. Differentiation in r, followed by integration by parts, gives

$$r\partial_r H_r(R) = (1 + e^{-rR})\,\mathrm{Ein}(R) + 2\,\mathrm{Ein}(rR) - 2\,\mathrm{Ein}((r+1)R). \tag{3.10}$$

Using $\mathrm{Ein}(R) = \gamma + \log R + E_1(R)$ and $H_1 = 0$, integration in r yields the locally uniform limit

$$\lim_{R\to\infty} \{H_r(R) - \log r \log R\} = \gamma \log r - 2\,\mathrm{Li}_2(-1/r) - \frac{\pi^2}{6}. \tag{3.11}$$

The standard exponential-integral identity is in [23]. This limit connects the iterated integral to familiar constants, but neither transcendence at finite algebraic points nor algebraic independence is automatically preserved by this limiting operation. Even transcendence of the displayed limit would not by itself isolate γ from the dilogarithm and π^2 terms.

4 Gaussian endpoint separation and finite quantile classification

For a positive integer d put

$$J_d(x) = \int_0^x e^{-t^{2d}}\,\mathrm{d}t.$$

This is odd and strictly increasing on $\mathbb{R}$. To use E-functions one must not simply call $J_d(z)$ an E-function. The appropriate reparametrization is

$$E_{c,d}(z) = \int_0^1 e^{-czt^{2d}}\,\mathrm{d}t = \sum_{n\ge 0} \frac{(-c)^n}{2dn+1}\frac{z^n}{n!}, \qquad c > 0. \tag{4.1}$$

It satisfies

$$2dzE'_{c,d} + E_{c,d} = e^{-cz}. \tag{4.2}$$

For algebraic c, conjugates of its normalized coefficients are exponentially bounded. If Mc is integral, a common denominator through order n divides $M^n \operatorname{lcm}(1, \dots, 2dn+1)$, which has exponential size.

Lemma 4.1. *For distinct positive algebraic $r_1, \dots, r_s$, the numbers*

$$1,\ J_d(r_1), \dots, J_d(r_s),\ e^{-r_1^{2d}}, \dots, e^{-r_s^{2d}}$$

are linearly independent over $\overline{\mathbb{Q}}$.

Proof. We first prove functional linear independence of $1, E_{c_i,d}, e^{-c_i z}$ for $0 < c_1 < \cdots < c_s$. In a hypothetical rational-function relation, take the largest c that occurs and let $z = -x \to -\infty$. If its $E_{c,d}$ coefficient is zero, the largest exponential cannot cancel. Otherwise the relation produces a rational R such that both $T = e^{cz}E_{c,d} - R$ and T' decay exponentially, up to polynomial factors. By (4.2), the rational function

$$1 - \{2dzR' + (1 - 2dcz)R\}$$

is exponentially small and therefore identically zero.

But $2dzR' + (1 - 2dcz)R = 1$ has no rational solution. At a finite nonzero pole, the derivative raises the pole order. At a pole of order a at zero the leading coefficient is multiplied by $1 - 2da \neq 0$. Thus R would be a polynomial, and then $-2dczR$ has an uncancellable leading degree.

The functions satisfy a finite E-system with only the finite singularity 0. Beukers's linear specialization theorem at 1, and $J_d(r) = rE_{r^{2d},d}(1)$, prove the assertion. □

4.1 The endpoint-vector algorithm

Let all data belong to a real number field K, and let

$$f_d(x) = \sum_{j=1}^{m} w_j e^{-[\lambda_j(x-\mu_j)]^{2d}}, \qquad a \le x \le b, \tag{4.3}$$

where $a < b$ and $w_j, \lambda_j > 0$. Merge identical (λ_j, μ_j). A primitive is $S_d(x) = \sum_j (w_j/\lambda_j) J_d(\lambda_j(x-\mu_j))$. For each positive radius r define

$$V_r(x) = \sum_{j:\,|\lambda_j(x-\mu_j)|=r} \frac{w_j}{\lambda_j} \operatorname{sgn}(x - \mu_j), \tag{4.4}$$

omitting zero radii. By Lemma 4.1, for algebraic x, q,

$$\frac{S_d(x) - S_d(a)}{S_d(b) - S_d(a)} = q \quad \Longleftrightarrow \quad V(x) - (1-q)V(a) - qV(b) = 0. \tag{4.5}$$

The right side contains no transcendental values and is independent of d.

Theorem 4.2. *All algebraic location–probability pairs in the interior of* (4.3) *lie in K^2 and can be found by inspecting at most $5m^2$ locations. Their set is independent of the positive integer d.*

Proof. Let $\mathcal{R}$ contain all positive endpoint radii $|\lambda_j(a-\mu_j)|$ and $|\lambda_j(b-\mu_j)|$; then $|\mathcal{R}| \le 2m$. The possible locations are the centers μ_j, the points $\mu_j \pm r/\lambda_j$ for $r \in \mathcal{R}$, and the solutions of

$$\lambda_j(x - \mu_j) = \varepsilon\lambda_k(x - \mu_k), \quad j < k, \quad \varepsilon = \pm 1,$$

with nonzero denominator. Their number is at most $m + 4m^2 + 2\binom{m}{2} = 5m^2$. Outside this set every moving radius is nonzero, distinct from the others, and distinct from the fixed endpoint radii. Its coefficient in (4.5) cannot vanish.

Every candidate lies in K. Set $D = V(b) - V(a) \neq 0$, the inequality following from positivity of the total mass. A coordinate $D_r \neq 0$ determines the only possible value $q = (V_r(x) - V_r(a))/D_r \in K$. Test all remaining coordinates and $0 < q < 1$. This also proves finiteness and independence of d. The bound counts candidates, not bit operations. □

4.2 Two explicit transcendental medians and genuine exceptions

For

$$f(x) = e^{-(x+1/2)^2} + 2e^{-(x-1/2)^2}, \qquad -1 \le x \le 1,$$

write $S(x) = J_1(x + 1/2) + 2J_1(x - 1/2)$. The quantile equation becomes

$$S(x) = (-1 + 3q)J_1(1/2) + (-2 + 3q)J_1(3/2).$$

For $-1 < x < 1$ no moving radius equals $3/2$. Thus an algebraic pair would require $q = 2/3$. The remaining candidate locations are $-1/2, 0, 1/2$; their residuals are respectively $-2J_1(1) - J_1(1/2)$, $-2J_1(1/2)$, and $J_1(1) - J_1(1/2)$, none zero. All algebraic-probability quantiles of this distribution are therefore transcendental, including its median

$$0.111102019216173792870452303270482046331 2\ldots.$$

Likewise the solution of $J_1(m_0) = J_1(1)/2$ is transcendental and equals

$$m_0 = 0.392697625792947214547243123745019495261 8\ldots.$$

On $[-3/2, 3/2]$, however, the same asymmetric density has the exact pair $(x, q) = (1/2, 2/3)$, and the finite criterion shows it is the only interior algebraic pair. A symmetric mixture on $[-1, 1]$ has $(0, 1/2)$; a single Gaussian on $[0, 1]$ has none. These examples prevent an unjustified statement that all asymmetric distributions have exclusively transcendental quantiles.

4.3 Why this does not settle a specified standard normal quantile

For the untruncated standard normal distribution,

$$\Phi(x) = \frac{1}{2} + \frac{J_1(x/\sqrt{2})}{\sqrt{\pi}}.$$

Two distinct positive algebraic quantiles at algebraic probabilities would contradict Lemma 4.1 after eliminating $\sqrt{\pi}$. Thus there is at most one such positive algebraic quantile. This does not identify whether a specified value such as $\Phi^{-1}(0.975)$ is that possible exception. The normalization at infinity is a real obstruction to this particular proof.

5 Logarithmic forms and statistical quantiles

5.1 A parity classification for Student distributions

Let $Q_\nu(q)$ be the exact quantile of the standard Student distribution with positive integral degrees of freedom ν. The distribution definitions and conventional tabulated critical values may be found in [24].

Theorem 5.1. *For $q \in \overline{\mathbb{Q}} \cap (0,1)$ the following classification holds.*

(i) If ν is even, $Q_\nu(q)$ is algebraic.
(ii) If $\nu \geq 3$ is odd, $Q_\nu(q)$ is transcendental unless $q = 1/2$, in which case it is zero.
(iii) If $\nu = 1$, the quantile is algebraic exactly when q is rational.

Proof. For $\nu = 2m+1 \geq 3$, let $u = x/\sqrt{\nu}$. Reduction of the rational integral gives

$$F_{2m+1}(x) = \frac{1}{2} + \frac{\arctan u + R_m(u)}{\pi}, \tag{5.1}$$

$$R_m(u) = \sum_{k=1}^{m} \frac{(2k-2)!!}{(2k-1)!!} \frac{u}{(1+u^2)^k}. \tag{5.2}$$

Every coefficient is positive, so $R_m(u) \neq 0$ for real $u \neq 0$. If x, q were both algebraic, the quantile equation would give

$$\log \frac{1+\mathrm{i}u}{1-\mathrm{i}u} - (2q-1)\log(-1) = -2\mathrm{i}R_m(u),$$

with $\log(-1) = \mathrm{i}\pi$ and the first logarithm equal to $2\mathrm{i}\arctan u$. The left side is a logarithmic form, whereas the right side is a nonzero algebraic number. Baker's theorem excludes this equality.

For $\nu = 2m$, set $y = x/\sqrt{2m + x^2}$. Then

$$F_{2m}(x) = \frac{1}{2} + c_m \sum_{k=0}^{m-1} (-1)^k \binom{m-1}{k} \frac{y^{2k+1}}{2k+1}, \qquad c_m = \frac{(2m)!}{4^m m!(m-1)!} \in \mathbb{Q}. \tag{5.3}$$

A specified algebraic value of this nonconstant polynomial forces y, and hence $x = \sqrt{2m}\, y/\sqrt{1-y^2}$, to be algebraic.

Finally $Q_1(q) = \tan(\pi(q - 1/2))$. Rational q gives a root-of-unity expression. For algebraic irrational q, algebraicity of the tangent would make $(-1)^{2q-1}$ algebraic, contrary to Gelfond–Schneider. □

Example 5.2. At $q = 39/40$, one has

$$Q_2(q) = 19\sqrt{2/39}, \qquad Q_3(q) = 3.18244630528370959272322542577986837626 89\ldots.$$

The second number is transcendental. Here Hermite–Lindemann alone suffices: if $u = Q_3(q)/\sqrt{3}$, $r = u/(1+u^2)$, and $A = (1+\mathrm{i}u)/(1-\mathrm{i}u)$ were algebraic, then

$$\arctan u + r = 19\pi/40 \quad \Longrightarrow \quad A^{20} e^{40\mathrm{i}r} = -1,$$

an impossibility. The even-degree example $Q_4(q)$ instead satisfies

$$39X^6 + 468X^4 - 2928X^2 - 23104 = 0.$$

These statements concern the exact values, not the rounded entries in a table.

5.2 Half-integral beta laws and three possible algebraic pairs

For integers $n, m \geq 0$, write

$$B_{n,m}(x) = \frac{1}{\mathrm{B}(n+1/2, m+1/2)} \int_0^x t^{n-1/2}(1-t)^{m-1/2}\,\mathrm{d}t.$$

When $n + m \geq 1$, set

$$K_{n,m} = \frac{4^{n+m} n!m!(n+m)!}{(2n)!(2m)!}.$$

There is a unique $P_{n,m} \in \mathbb{Q}[x]$ of degree at most $n + m - 1$ satisfying

$$2x(1-x)P' + (1-2x)P = 2K_{n,m}x^n(1-x)^m - 2. \tag{5.4}$$

Indeed, the operator on the left is injective, raises the degree by one, and has codimension-one image. Its image is the kernel of integration against $[2\sqrt{x(1-x)}]^{-1}$. The right side has zero such integral by normalization. Differentiating and checking the value at zero proves

$$B_{n,m}(x) = \frac{2\arcsin\sqrt{x} + \sqrt{x(1-x)}P_{n,m}(x)}{\pi}. \tag{5.5}$$

Theorem 5.3. *If $n + m \geq 1$, and algebraic $x, q \in (0,1)$ satisfy $B_{n,m}(x) = q$, then*

$$(q,x) \in \{(1/3,1/4), (1/2,1/2), (2/3,3/4)\}. \tag{5.6}$$

For $n, m \geq 1$, at most one of these pairs occurs. A listed pair occurs exactly when $P_{n,m}(x) = 0$. If exactly one of n, m is zero, none occurs.

Proof. For algebraic x, put $\theta = \arcsin\sqrt{x}$ and $A = (1 + \mathrm{i}\sqrt{x/(1-x)})/(1 - \mathrm{i}\sqrt{x/(1-x)})$. Equation (5.5) implies

$$q\log(-1) - \log A = \mathrm{i}\sqrt{x(1-x)}P_{n,m}(x).$$

Baker's theorem forces $P_{n,m}(x) = 0$. Then $A = (-1)^q$ is algebraic, so Gelfond–Schneider forces $q \in \mathbb{Q}$.

For $n, m \geq 1$, let $H(x) = B_{n,m}(x) - 2\arcsin\sqrt{x}/\pi$. Its derivative is

$$H'(x) = \frac{K_{n,m}x^n(1-x)^m - 1}{\pi\sqrt{x(1-x)}}.$$

The numerator changes sign in the order $-, +, -$. Its positive maximum exceeds zero because the corresponding density ratio has weighted mean one. Since $H(0) = H(1) = 0$, the function has exactly one zero in $(0,1)$. Consequently $P_{n,m}$ has exactly one root there.

As q is rational, $x = (2 - \zeta - \zeta^{-1})/4$ for the root of unity $\zeta = e^{\mathrm{i}\pi q}$. All algebraic conjugates of x lie strictly in $(0,1)$, and all are roots of the rational polynomial $P_{n,m}$. Uniqueness of its real interior root therefore forces $x \in \mathbb{Q}$. Now $2\cos(\pi q) = 2 - 4x$ is a rational algebraic integer in $(-2,2)$, hence is $-1, 0$, or 1. This gives (5.6).

If $n = 0 < m$, the density ratio $K_{0,m}(1-x)^m$ is strictly decreasing, and H is positive in the interior; for $m = 0 < n$ it is negative. Thus no root occurs. Conversely, every pair in (5.6) has $2\arcsin\sqrt{x} = \pi q$, so $P(x) = 0$ is sufficient. □

For $n = m = 0$ the law is the arcsine distribution; its algebraic-probability quantile is algebraic exactly when the probability is rational. This excluded case cannot be folded into Theorem 5.3.

A practical exact recurrence follows by writing the right side of (5.4) as $\sum h_k x^k$ and $P = \sum a_k x^k$:

$$a_{-1} = 0, \qquad a_k = \frac{h_k + 2ka_{k-1}}{2k+1}, \qquad -2(n+m)a_{n+m-1} = h_{n+m}.$$

This tests the three candidate locations for each specified parameter pair. It does not classify all integer pairs (n,m) for which $P_{n,m}(1/4) = 0$.

Example 5.4. For $\mathrm{Beta}(3/2, 5/2)$,

$$P_{1,2}(x) = -\frac{2}{3}(4x-1)(2x-3).$$

The only interior algebraic pair is $(q, x) = (1/3, 1/4)$. The median and both 95% equal-tailed endpoints are transcendental:

$$\begin{aligned} Q(1/2) &= 0.352452279877130813765599743574394051817 8\ldots, \\ Q(1/40) &= 0.038747617785165160780701942186546689742 7\ldots, \\ Q(39/40) &= 0.823263902868742682562881188291525574485 3\ldots. \end{aligned}$$

5.3 Binomial intervals and the F distribution

With s successes in $N \geq 1$ Bernoulli observations, the Jeffreys posterior is $\mathrm{Beta}(s+1/2, N-s+1/2)$. Thus its unmodified 95% equal-tailed posterior interval has two transcendental endpoints, since $1/40, 39/40$ are absent from (5.6). Replacing an endpoint by 0 or 1 in a boundary-adjusted implementation is a different definition and of course gives an algebraic replacement. In contrast, the Clopper–Pearson endpoints use integer beta parameters and rational probabilities; their distribution functions are rational polynomials, so all exact endpoints are algebraic. The statistical definitions are described in [25]. This arithmetic distinction asserts no coverage or performance superiority.

For Y with an $F(d_1, d_2)$ law, $X = d_1Y/(d_1Y + d_2)$ has beta parameters $(d_1/2, d_2/2)$, and the inverse transformation is rational. If one degree of freedom is even, all algebraic-probability quantiles are algebraic, by a polynomial integral after a square-root substitution if necessary. If both are odd and not both one, there is at most one interior algebraic pair, at a probability in $\{1/3, 1/2, 2/3\}$. In particular every such 95% quantile is transcendental. For $F(3, 5)$ the sole exception is $Q(1/3) = 5/9$.

6 Shortest beta probability intervals

The previous results prescribe a single probability $F(x)$. A shortest interval instead imposes two simultaneous equations, $F(b) - F(a) = c$ and $f(a) = f(b)$; neither $F(a)$ nor $F(b)$ need be algebraic. This requires a new noncancellation argument.

Theorem 6.1. *Let $\alpha, \beta \in \mathbb{Q}$ satisfy $\alpha, \beta > 1$ and $\alpha + \beta \in \mathbb{Z}$. For algebraic $c \in (0, 1)$, let $[a, b]$ be the unique shortest interval carrying beta probability c, measured in the original x coordinate. If the parameters are integers, $a, b, b - a$ are algebraic. Otherwise each is transcendental, and*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(a, b) = 1.$$

In the nonintegral case the midpoint is transcendental unless $\alpha = \beta$, when it is $1/2$.

6.1 Existence, uniqueness, and the density constraint

For $\alpha, \beta > 1$, the beta density is zero at both ends, strictly increasing up to its mode, and then strictly decreasing. Its positive superlevel sets are intervals $I_\lambda = [a_\lambda, b_\lambda]$ whose mass runs continuously and strictly from zero to one. Thus the interval of any specified mass exists uniquely and has equal endpoint densities. If J were shorter than I_λ, then

$$\int_{I_\lambda} f - \int_J f \geq \lambda\big(|I_\lambda \setminus J| - |J \setminus I_\lambda|\big) = \lambda(|I_\lambda| - |J|) > 0.$$

This proves optimality, and strict unimodality gives uniqueness. It is the usual highest-density interval, not an equal-tailed interval; the endpoint condition also appears in [26].

6.2 A monotone correction divided by the density

For nonintegral parameters write

$$\alpha = n + r, \quad \beta = m + 1 - r, \quad n, m \geq 1, \quad 0 < r < 1, \qquad r \in \mathbb{Q}.$$

Set $M = n + m$ and

$$w_0 = x^{r-1}(1-x)^{-r}, \quad B_0 = \mathrm{B}(r, 1-r) = \frac{\pi}{\sin \pi r}, \quad K = \frac{M!}{(r)_n (1-r)_m}.$$

There is a unique rational polynomial P of degree at most $M - 1$ with

$$x(1-x)P' + (r-x)P = Kx^n(1-x)^m - 1. \tag{6.1}$$

The same codimension-one weighted-integral argument as before proves this assertion. If $D = x^r(1-x)^{1-r}P$, then

$$B_0 F(x) = \int_0^x w_0(t)\,\mathrm{d}t + D(x). \tag{6.2}$$

Lemma 6.2. *The function $R(x) = P(x)/[x^{n-1}(1-x)^{m-1}]$ satisfies $R'(x) > 0$ on $(0,1)$. Consequently $f(a) = f(b) > 0$ and $a < b$ imply $D(b) - D(a) > 0$.*

Proof. Write $P = \sum_{k=0}^{M-1} p_k x^k (1-x)^{M-1-k}$. With $p_{-1} = p_M = 0$, coefficient comparison in (6.1) yields

$$(j + r)p_j - (M - j + 1 - r)p_{j-1} = K\mathbf{1}_{j=n} - \binom{M}{j}.$$

The forward recurrence from $p_0 = -1/r$ gives $p_k < 0$ for $k < n$; the backward recurrence from $p_{M-1} = 1/(1-r)$ gives $p_k > 0$ for $k \geq n$. With $j = k - n + 1$, a term of R is $p_k x^j (1-x)^{1-j}$, whose derivative is

$$p_k x^{j-1}(1-x)^{-j}(j-x) > 0.$$

For $k < n$, both p_k and $j - x$ are negative; otherwise both are positive. Summing proves the assertion. Finally

$$D(x) = \frac{B_0}{K} f(x) R(x),$$

so equal positive densities imply strict increase of D between the two endpoints. □

6.3 Transcendence of endpoints, width, and midpoint

Suppose a, b were algebraic. By Lemma 6.2, $D(b) - D(a)$ is a nonzero algebraic number. But (6.2) gives

$$D(b) - D(a) = cB_0 - \int_a^b w_0(x)\,\mathrm{d}x.$$

Write $r = p/q$ in lowest terms and substitute $y = (x/(1-x))^{1/q}$. Then

$$w_0(x)\,\mathrm{d}x = \frac{q y^{p-1}}{1 + y^q}\,\mathrm{d}y.$$

This is a proper rational function with simple algebraic poles. Its integral between algebraic endpoints is an algebraic-coefficient logarithmic form. The same is true of cB_0, since $\sin \pi r$ is nonzero algebraic. Baker's theorem contradicts the nonzero algebraic left side.

Clearing denominators in the density equation gives

$$h(a) = h(b), \qquad h(X) = X^A(1-X)^B \in \mathbb{Q}[X], \quad A, B \in \mathbb{Z}_{>0}. \tag{6.3}$$

If one endpoint were algebraic, the other would also be algebraic. Thus each is transcendental, while their polynomial relation makes their joint transcendence degree exactly one.

If $w = b - a > 0$ were algebraic, a would solve the nonzero polynomial $h(X + w) - h(X)$ over $\overline{\mathbb{Q}}$. Its leading term is $d\kappa w X^{d-1}$ when κX^d is the leading term of h. This is impossible. If the midpoint $z = (a+b)/2$ were algebraic, the analogous polynomial $h(2z - X) - h(X)$ would have to vanish identically. Reflection must interchange the only roots 0, 1, preserving multiplicities, so $z = 1/2$ and $A = B$. This is exactly the symmetric case.

For integer parameters let $A = \alpha - 1$, $B = \beta - 1$. The equal-density curve has the rational parametrization

$$a(t) = \frac{t^A - 1}{t^{A+B} - 1}, \qquad b(t) = t^B a(t), \qquad t > 1.$$

The mass $F(b(t)) - F(a(t))$ is a nonconstant rational function, tending to zero and one at the two ends of this parameter range. At an algebraic mass, t and hence the endpoints and width are algebraic. This completes the proof of Theorem 6.1.

6.4 Examples and posterior interpretation

For $\mathrm{Beta}(3/2, 5/2)$ at mass $19/20$, the defining equations are

$$a(1-a)^3 = b(1-b)^3, \qquad \frac{16}{\pi}\int_a^b \sqrt{x}(1-x)^{3/2}\,\mathrm{d}x = \frac{19}{20}.$$

The endpoints and width are

$$\begin{aligned} a &= 0.00955687087304494286021945578899816354395\ldots, \\ b &= 0.77075726925733542833483601801490110700 37\ldots, \\ b - a &= 0.76120039838429048547461656222591947156 42\ldots. \end{aligned}$$

All three, and their nonsymmetric midpoint, are transcendental. In the parametrization $S = 1 + t + t^2 + t^3$, $a = 1/S$, $b = t^3/S$, one can independently check

$$D(b) - D(a) = \frac{2(t-1)\sqrt{t(t^2+t+1)}}{3S^3}(3t^6 + 14t^5 + 25t^4 + 36t^3 + 25t^2 + 14t + 3) > 0.$$

For the integer-parameter comparison $\mathrm{Beta}(2, 3)$, the 95% value of t instead solves

$$t^8 + 4t^7 + 10t^6 - 64t^5 - 361t^4 - 544t^3 - 390t^2 - 156t - 39 = 0.$$

If both successes and failures are at least one, a Jeffreys binomial posterior has nonintegral half-integer parameters of integral sum. Its shortest interval at every algebraic mass has transcendental endpoints and width. A uniform-prior posterior has integer parameters, and its corresponding quantities are algebraic. The conclusion is coordinate dependent: a shortest interval after a nonlinear reparametrization is a different optimization problem. Boundary data, U-shaped laws, and arbitrary rational parameter sums are not covered by this theorem.

7 Polynomial lemniscates: lengths as closed periods

For nonconstant $P \in \overline{\mathbb{Q}}[z]$ and algebraic $\rho > 0$, let

$$\Lambda_{P,\rho} = \{z \in \mathbb{C} : |P(z)| = \rho\}, \qquad L(P,\rho) = \text{length}(\Lambda_{P,\rho}).$$

Length is that of the geometric set; repeated polynomial factors do not multiply it. The argument below includes critical levels, crossing branches, and multiple connected components.

Theorem 7.1. *The number $L(P,\rho)$ is transcendental. Moreover,*

$$\frac{L(P,\rho)}{2\pi} \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad P \text{ has exactly one distinct zero.} \tag{7.1}$$

If P has at least two distinct zeros, then $1, \pi, L(P,\rho)$ are linearly independent over $\overline{\mathbb{Q}}$, and more generally $L(P,\rho)$ is not an algebraic number plus an algebraic linear combination of logarithms of algebraic numbers. In particular $e^{aL(P,\rho)}$ is transcendental for every $0 \neq a \in \overline{\mathbb{Q}}$.

This formulation extends the length question posed in [7, Remark 8.16]. It makes no assertion about which polynomial of fixed degree maximizes the length.

7.1 Removing powers and constructing one connected curve

Write $P/\rho = Q^d$ so that the multiplicities m_j of the distinct zeros of Q have greatest common divisor one. A dth root of the leading coefficient is algebraic, and the level set is unchanged. Put $n = \deg Q$. Consider the complex curve

$$Q(z)\overline{Q}(w) = 1, \tag{7.2}$$

where conjugation in $\overline{Q}$ acts on coefficients.

Lemma 7.2. *The curve* (7.2) *is geometrically irreducible.*

Proof. At a zero a_j of Q of multiplicity m_j, every root ξ of $\overline{Q}(w) - 1/Q(z)$ has valuation $-m_j/n$. Nonnegative valuation would be incompatible with the pole of $1/Q$; at negative valuation, the leading term of $\overline{Q}$ is uniquely minimal. If a monic factor of degree e existed over $\overline{\mathbb{Q}}(z)$, its nonzero constant coefficient, a product of e such roots, would have integral valuation $-em_j/n$. Thus $n \mid em_j$ for every j, and the greatest-common-divisor condition forces $n \mid e$. There is no proper factor. Gauss's lemma gives geometric irreducibility over the algebraically closed coefficient field. □

Related irreducibility statements occur in Orevkov [8]; the valuation argument is included to specify the connected model used here. Let X be its smooth projective normalization, with real structure $(z,w) \mapsto (\overline{w},\overline{z})$. Set $x = (z+w)/2$, $y = (z-w)/(2\mathrm{i})$. The real affine part is the original level set. Its highest homogeneous real term is $|a_n|^2(x^2+y^2)^n$, so it has no real point at infinity.

7.2 Critical points are crossings, not cusps

At a point z_0 of the level set, $Q(z_0) \neq 0$, so locally

$$g(z) = \log \frac{Q(z)}{Q(z_0)}$$

is holomorphic. If its zero has order m, choose a local holomorphic coordinate u with nonzero derivative so that $g = u^m$. The real level set is then $\mathrm{Re}(u^m) = 0$, the union of m distinct straight lines. Each line is a

smooth immersed branch in the z plane after reversing the coordinate change. Normalization separates these branches. Hence on every real branch of X,

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 > 0. \tag{7.3}$$

The compact real locus of X is a finite disjoint union of circles. It is nonempty, since a noncritical value of Q on the unit circle has a regular inverse branch. The finite collection of crossing points has length zero, so summing the lengths of the normalized circles gives precisely the length of the geometric level set.

7.3 The positive square root closes on an algebraic cover

On X set $q = dz\,dw = (dx)^2 + (dy)^2$. Choose an algebraic nonzero rational one-form τ and write $q = f\tau^2$. If f is not a square, take the connected degree-two cover $\pi : Y \to X$ with function field obtained by adjoining $v^2 = f$, and put $\omega = v\pi^*\tau$. If f is a square, use $Y = X$ and a global square root. In either case

$$\omega^2 = \pi^* q. \tag{7.4}$$

The square root alone does not prove that the length is a closed period. On an oriented real circle $C \subset X(\mathbb{R})$, choose at each point the sheet on which ω evaluates positively on the tangent direction. By (7.3) the cover is unramified there, and this choice is unique and continuous. The point and oriented tangent agree after a complete turn, so the chosen sheet also agrees. Thus the lifted path $\widetilde{C}$ is closed and

$$\int_{\widetilde{C}} \omega = \text{length}(C \text{ in the plane}) > 0.$$

If the square root is global, reverse the orientation of each circle when necessary. All circles can then be combined on the same connected Y into an integral cycle γ satisfying

$$\partial\gamma = 0, \qquad \int_\gamma \omega = L(P, \rho) > 0. \tag{7.5}$$

The closed-period theorem quoted in Section 2 proves transcendence. This is one nonzero period, not an unsupported sum-of-transcendentals argument. No limiting passage from regular to singular levels is used.

7.4 Global holomorphy and the exact circle exception

Suppose Q has at least two distinct zeros. The differential $q = dz\,dw$ has no pole where z, w are finite. At an infinite point, exactly one of z, w tends to infinity and the other approaches a root of the other polynomial. Suppose $z = \infty$, $w = a$, where a has multiplicity $m < n$. With $d = \gcd(n, m)$, normalization gives a local parameter t with

$$z \sim c\,t^{-m/d}, \qquad w - a \sim c'\,t^{n/d}.$$

Consequently

$$\text{ord}_t(q) = \frac{n-m}{d} - 2 \geq -1. \tag{7.6}$$

The same holds with z, w interchanged. If locally $q = t^k h(t)(dt)^2$, $h(0) \neq 0$, then for even k the square root has order $k/2 \geq 0$. For odd k, the substitution $t = u^2$ makes its order $k + 1 \geq 0$. Thus ω is globally holomorphic on Y.

The period (7.5) is now a coordinate of an abelian logarithm of the identity in $\text{Jac}(Y)$. Apply Lemma 2.4 to obtain the stronger logarithmic separation in Theorem 7.1. If $P(z) = c(z-a)^n$, the level set is a circle of algebraic radius $(\rho/|c|)^{1/n}$ and its length is 2π times that radius. In this case $m = n$ in the order calculation, producing a double pole of q and a simple pole of its square root. This proves the precise dichotomy and all stated consequences.

7.5 A necessary counterexample outside the hypothesis

The algebraic curve parametrized by $z(t) = (1 + e^{it})^2$, $-\pi \le t \le \pi$, has speed $4\cos(t/2)$ and total length 16. It has a cusp where the normalization map is not an immersion. A positive square-root branch cannot be continued through that point as the same closed holomorphic lift. This prevents an incorrect extension of Theorem 7.1 to every closed algebraic curve.

7.6 A convergent certificate for a concrete normalized length

Let $L_3 = L(z^3 - z, 1)$. Define $y(t)$ by $y^3 - ty = 1$, $y(0) = 1$, and

$$G(t) = \sqrt{\frac{3}{3y(t)^2 - t}} = \sum_{n\ge 0} g_n t^n, \qquad G(0) = 1.$$

All g_n are rational. The exterior parametrization $\phi(u) = uy(u^{-2})$ satisfies $\phi^3 - \phi = u^3$ and parametrizes the level curve once on $|u| = 1$. Its derivative is $G(u^{-2})^2$. Fourier orthogonality therefore gives

$$\frac{L_3}{2\pi} = \sum_{n\ge 0} g_n^2. \tag{7.7}$$

The coefficients can be computed without numerical root finding:

$$y_0 = 1, \quad y_n = \frac{(-1)^{n-1}}{3n}\binom{(2n-4)/3}{n-1}, \quad d_n = (1-2n)y_n,$$

$$g_0 = 1, \qquad g_n = \frac{1}{2}\left(d_n - \sum_{k=1}^{n-1} g_k g_{n-k}\right).$$

On $|t| \le 3/2$, all roots of $y^3 - ty - 1$ have modulus less than $3/2$. Since

$$\prod_{j=1}^{3}(3y_j^2 - t) = 27 - 4t^3,$$

we obtain $|3y_j^2 - t| \ge 24/121$ and $|G(t)|^2 \le 121/8$. Cauchy's estimate gives the explicit tail bound

$$\sum_{n>M} g_n^2 \le \frac{1089}{40}\left(\frac{4}{9}\right)^{M+1}. \tag{7.8}$$

The values are

$$L_3 = 6.464706327813477265623126732842916032783 0\ldots,$$
$$L_3/\pi = 2.057779935417939294967383271828091319132 5\ldots.$$

Both are transcendental, and $1, \pi, L_3$ are linearly independent. Algebraic independence of π and L_3 is not asserted.

8 Cyclotomic homology and all rational arc divisions

For every positive integer N, put

$$I_N(s) = \int_0^s \frac{dt}{\sqrt{1-t^N}}, \qquad R_N(s) = \frac{I_N(s)}{I_N(1)}, \qquad 0 \le s \le 1. \tag{8.1}$$

Let $s_{N,q}$ be the unique solution of $R_N(s) = q$ in $(0, 1)$.

Theorem 8.1. *For every rational $q \in (0,1)$,*

$$s_{N,q} \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad N \in \{1,2,3,4,6\}.$$

For every other positive integral exponent, each of these division values is transcendental.

The distinction between the exponent N and a lemniscate's number of leaves is important: a curve with n leaves uses $N = 2n$.

8.1 The primitive cyclotomic factor

For $N \geq 3$ consider $\mathcal{H}_N : y^2 = 1 - x^N$ and its Jacobian J_N. Its genus is $g = \lfloor (N-1)/2 \rfloor$, with holomorphic basis

$$\omega_k = x^{k-1}dx/y, \qquad 1 \leq k \leq g.$$

The rotation $T : (x, y) \mapsto (\zeta_N x, y)$ has eigenvalue ζ_N^k on ω_k. Its characteristic polynomial on rational homology is

$$\chi_N(X) = \begin{cases} (X^N - 1)/(X - 1), & N \text{ odd}, \\ (X^N - 1)/(X^2 - 1), & N \text{ even}. \end{cases} \tag{8.2}$$

All eigenvalues occur once. A rational polynomial idempotent e_N selects the Φ_N factor:

$$e_N \equiv 1 \pmod{\Phi_N}, \qquad e_N \equiv 0 \pmod{\chi_N/\Phi_N}.$$

After clearing denominators by $d > 0$, set $A_N = \operatorname{im}(de_N(T))$ and $\pi : J_N \to A_N$. This standard construction is explained in [9]. It gives

$$H_1(A_N, \mathbb{Q}) \simeq \mathbb{Q}[X]/(\Phi_N(X)), \qquad \dim A_N = \varphi(N)/2. \tag{8.3}$$

For every $1 \leq k < N/2$ with $\gcd(k, N) = 1$, there is an algebraic invariant form η_k with $\pi^*\eta_k = \omega_k$. Explicitly, if $i : A_N \hookrightarrow J_N$, take $\eta_k = d^{-1}i^*\omega_k$. The rational action is irreducible, but A_N need not be a simple abelian variety. This distinction permits $N = 8$ and composite exponents.

Lemma 8.2 (Injective rational periods)**.** *The period map $\mathcal{P} : H_1(A_N, \mathbb{Q}) \to \mathbb{C}$, $\gamma \mapsto \int_\gamma \eta_1$, is injective. If U is a logarithm of an algebraic point of A_N and $\eta_1(U) = 0$, then $U = 0$.*

Proof. As $\mathcal{P}(T_*\gamma) = \zeta_N \mathcal{P}(\gamma)$, its kernel is a rational T-invariant subspace. Irreducibility of (8.3) makes the kernel zero or the whole space. In the latter case the nonzero invariant form vanishes on the entire period lattice, which spans the real Lie algebra, an impossibility.

Apply External theorem 2.3 to U and $\ker \eta_1$. A positive-dimensional abelian subgroup there has a nonzero rational cycle, injective in the homology of A_N, with zero period. The first assertion excludes this. Thus the subgroup, and then U, is zero. □

8.2 A second integral contradicts proportional division

If $\varphi(N) \geq 4$, there exists $1 < k < N/2$ with $\gcd(k, N) = 1$. On the positive real path $P(t) = (t, \sqrt{1 - t^N})$, based at $O = (0, 1)$, let $V(s)$ be the actual Abel logarithm projected to A_N. Then

$$\exp_{A_N} V(s) = \pi([P(s) - O]), \qquad \eta_j(V(s)) = I_{N,j}(s) := \int_0^s \frac{t^{j-1}\,\mathrm{d}t}{\sqrt{1 - t^N}}.$$

All these integrals converge at 1. If $q = a/b \in \mathbb{Q} \cap (0, 1)$ and $s = s_{N,q}$ were algebraic, the vector $U = bV(s) - aV(1)$ would exponentiate to an algebraic point and would satisfy $\eta_1(U) = 0$. By Lemma 8.2, $U = 0$.

But with $w_N(t) = (1 - t^N)^{-1/2}$, the weighted mean

$$M_k(s) = \frac{I_{N,k}(s)}{I_N(s)}$$

is strictly increasing, since

$$M_k'(s) = \frac{w_N(s)}{I_N(s)^2} \int_0^s (s^{k-1} - t^{k-1}) w_N(t)\, \mathrm{d}t > 0. \tag{8.4}$$

It follows that

$$\eta_k(U) = bI_N(s)\{M_k(s) - M_k(1)\} < 0,$$

a contradiction. This proves the transcendental cases of Theorem 8.1.

8.3 The small-exponent algebraic cases

For $N = 1$, $s_{1,q} = 2q - q^2$. For $N = 2$, $s_{2,q} = \sin(\pi q/2)$, algebraic at rational q. For $N = 3, 4$, the curve has genus one and the endpoint divisor $[W - O]$, $W = (1, 0)$, is torsion. For $N = 3$, the point O is 3-torsion relative to infinity, since $y = 1$ meets the cubic there with multiplicity three, while W is 2-torsion. For $N = 4$, taking O as origin, $(x, y) \mapsto (-x, y)$ is negation, $O' = (0, -1)$ is 2-torsion, and the hyperelliptic involution implies $2W = O'$. Thus W has order dividing four.

For $N = 6$, the finite map

$$(x, y) \longmapsto (u, v) = (x^2, xy), \qquad v^2 = u - u^4, \qquad du/v = 2dx/y$$

lands on a genus-one curve and sends both endpoints to branch points. Their difference is 2-torsion. A rational multiple of a torsion logarithm exponentiates to a torsion point. Such points on an algebraic elliptic curve, and preimages under a finite algebraic map, are algebraic. These observations prove all exceptional cases. They agree with the classical small-leaf results reviewed in [7].

8.4 The prime-only route and why it was superseded

For a prime $p \geq 5$, the curve $y^2 = x - x^{p+1}$ is isomorphic over $\overline{\mathbb{Q}}$ to $v^2 = 1 - u^p$, by $u = 1/x$, $v = \mathrm{i}y/x^{(p+1)/2}$. Its Jacobian is absolutely simple, as in [7, Proposition 4.8]. Along $(x, y) = (s^2, s\sqrt{1 - s^{2p}})$, the first two holomorphic forms pull back to $2\,\mathrm{d}s/\sqrt{1 - s^{2p}}$ and $2s^2\,\mathrm{d}s/\sqrt{1 - s^{2p}}$. Simplicity and the analytic subgroup theorem then give the same weighted-mean contradiction. This was the prime-leaf argument. The rational-period proof above removes the unnecessary simplicity hypothesis and includes all composite exponents.

For $N = 8$ the projector is particularly explicit:

$$\chi_8 = (X^2 + 1)(X^4 + 1), \qquad e_8(T) = \frac{1 - T^4}{2}.$$

Its two-dimensional image uses dx/y and x^2dx/y. This is compatible with a Jacobian isogenous to a product of elliptic curves: irreducibility of a rational rotation representation is not absolute simplicity.

8.5 Geometric division points and their boundary exceptions

The n-leaf curve $C_n = \{|z^n - 1| = 1\}$ has polar equation $r^n = 2\cos(n\theta)$ on the leaf about the positive real axis. Its half-leaf arclength from the origin is

$$\ell(r) = 2^{1/n} I_{2n}(r/2^{1/n}). \tag{8.5}$$

Thus every interior rational division point on a half-leaf is transcendental in its normalized radius when $n \geq 4$. Both Cartesian coordinates are individually transcendental: if one were algebraic, the equation

$$(X^2 + Y^2)^n - 2\,\mathrm{Re}(X + \mathrm{i}Y)^n = 0$$

is monic of degree $2n$ in the other coordinate, forcing that coordinate and then the radius to be algebraic.

The origin and leaf tips remain algebraic. In a traversal starting at the origin and visiting each leaf once, the jth point of an M-fold equal arclength division lies at a half-leaf boundary exactly when $M \mid 2nj$. Hence, for $n \geq 4$, all M-division points are algebraic exactly when $M \mid 2n$. Several boundary times can represent the same origin; this is a statement about the specified traversal, not a count of distinct plane points.

The first new normalized radii include

$$\begin{aligned} s_{8,1/2} &= 0.581372564921331418342397202120347148670 7\ldots, \\ s_{10,1/2} &= 0.566067311893637997084837911607046070621 7\ldots, \\ s_{12,1/2} &= 0.555521188888391003316451341835\ldots. \end{aligned}$$

For algebraic $0 \leq a < b \leq 1$, the same proof based at $P(a)$ shows that rational division of $\int_a^b w_N(t)\,\mathrm{d}t$ gives a transcendental interior location whenever $\varphi(N) \geq 4$. Finally,

$$I_N(s) = \frac{1}{N} B_{s^N}(1/N, 1/2),$$

so the rational quantiles of $\mathrm{Beta}(1/N, 1/2)$ are algebraic exactly for $N = 1, 2, 3, 4, 6$; in all other cases they are transcendental.

9 Algebraic division ratios and torsion

Rational division uses integer multiplication of logarithms. Algebraic irrational multiplication is not automatically an algebraic-group operation. The missing step is supplied by the period field.

Lemma 9.1 (An algebraic ratio comes from an endomorphism)**.** *Let $A/\overline{\mathbb{Q}}$ have a finite-order automorphism T whose characteristic polynomial on rational homology is Φ_N. Set $K = \mathbb{Q}[T] \simeq \mathbb{Q}(\zeta_N)$. Let $\eta \neq 0$ be an algebraic eigenform, with eigenvalue corresponding to the embedding $\sigma : K \hookrightarrow \mathbb{C}$. If u, v are logarithms of algebraic points, $\eta(v) \neq 0$, and $\lambda = \eta(u)/\eta(v)$ is algebraic, then $\lambda \in \sigma(K)$ and*

$$u = da(v), \qquad a \in K, \quad \sigma(a) = \lambda. \tag{9.1}$$

If $\exp_A(v)$ is torsion, algebraicity of this ratio is equivalent to torsion of $\exp_A(u)$.

Proof. As in Lemma 8.2, the period map is injective. Since its domain is a rank-one K-module, its image is $\Omega\sigma(K)$ for some $\Omega \neq 0$. Apply External theorem 2.3 to $A \times A$, (u, v), and the hyperplane $\eta(Z_1) - \lambda\eta(Z_2) = 0$. The resulting subgroup has a positive-dimensional second projection, because $v \neq 0$. Choose a rational cycle with nonzero second projection. Its second period is nonzero by injectivity, and integration of the hyperplane relation expresses λ as the ratio of two elements of $\Omega\sigma(K)$. Thus $\lambda \in \sigma(K)$.

Only now select the corresponding $a \in K \subseteq \mathrm{End}^0(A)$. Clearing denominators, $Mu - d(Ma)(v)$ exponentiates to an algebraic point and is annihilated by η. The last assertion of Lemma 8.2 makes it zero. This proves (9.1). A torsion logarithm is a rational period vector, and quasi-endomorphisms preserve rational periods. Conversely, the period-image description gives an algebraic ratio of two torsion logarithms. □

9.1 The reference endpoint is always torsion

On $\mathcal{H}_N$, put $O = (0, 1)$, $O' = (0, -1)$, and $W_r = (r, 0)$ for $r^N = 1$. The divisor identities

$$\operatorname{div} \frac{y-1}{y+1} = N(O - O'), \qquad \operatorname{div} \frac{x-r}{x} = 2W_r - O - O' \tag{9.2}$$

show $2N[W_r - O] = 0$. In particular $V(1)$ in Section 8 is a nonzero logarithm of a torsion point after projection. For algebraic $s \in (0, 1)$, Lemma 9.1 therefore gives the exact equivalence

$$R_N(s) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \pi([P(s) - O]) \text{ is torsion in } A_N. \tag{9.3}$$

The right side concerns the projected point. Replacing it by torsion in J_N without an additional argument would be invalid.

9.2 Complete exclusion for $N = 8, p, 2p$

Theorem 9.2. *Let $p \geq 5$ be prime and $N \in \{8, p, 2p\}$. Every algebraic $s \in (0, 1)$ has transcendental $R_N(s)$. Equivalently, every algebraic-probability inverse value $s_{N,q}$ is transcendental.*

Proof. For $N = 8$, use $C : y^2 = x - x^5$ with $P(s) = (s^2, s\sqrt{1 - s^8})$, $O = (0, 0)$, $W = (1, 0)$. Its genus-two Jacobian has the automorphism $(x, y) \mapsto (\mathrm{i}x, \zeta_8 y)$ and full homology characteristic polynomial Φ_8. Moreover $P^*(dx/y) = 2\,\mathrm{d}s/\sqrt{1 - s^8}$ and $[W - O]$ is 2-torsion. Thus an algebraic ratio forces $[P(s) - O]$ itself to be torsion.

Choose $\delta^2 = \zeta_8$ and set $X = \zeta_8 x$, $Y = \delta y$. The equation becomes $Y^2 = X^5 + X$. Boxall–Grant's complete packet [10, Theorem (0.3)] consists of infinity and the points whose X-coordinates are roots of

$$X^5 + X, \qquad X^4 + 4X^2 + 1, \qquad X^4 - 4X^2 + 1.$$

There are 22 points in total, including infinity. The change of base point from $(0, 0)$ to infinity adds 2-torsion. On $x = s^2 \in (0, 1)$ the three expressions become

$$\zeta_8 x(1 - x^4), \qquad 1 - x^4 + 4\mathrm{i}x^2, \qquad 1 - x^4 - 4\mathrm{i}x^2,$$

all nonzero. No point of our open arc belongs to the packet.

For $N = p$, the entire Jacobian of $y^2 = 1 - x^p$ is the primitive factor. With $X = -x$, $Y = y$ it becomes $Y^2 = X^p + 1$, and our arc has $-1 < X < 0$. The base point differs from infinity by torsion because $\operatorname{div}(y - 1) = p(O - \infty)$. For $N = 2p$, instead use $y^2 = x - x^{p+1}$ and the same squared-parameter path as above. Its full homology is the Φ_{2p} module. The transformation

$$X = -1/x, \qquad Y = \mathrm{i}y/x^{(p+1)/2}$$

sends it to $Y^2 = X^p + 1$, sends O to infinity, and sends our arc to $X < -1$.

Arul's classification [11, Theorem 6.2.21] says that for $p \geq 7$ only infinity or a zero coordinate can give torsion relative to infinity. For $p = 5$ the additional orbit has $X^5 = 4$. None lies on either negative real arc. This excludes torsion in both cases and proves the theorem. □

For example $R_8(1/2) = 0.42979706813511177972719300735477067271 40\ldots$ and $R_{10}(1/2) = 0.44159530366266713149330751734049766386 40\ldots$ are transcendental. This is stronger than their irrationality, which alone would follow by contraposition from rational-division results.

9.3 General algebraic exceptions: fields, conjugates, and finiteness

Suppose $\varphi(N) \geq 4$ and algebraic $s \in (0,1)$ has algebraic ratio $q = R_N(s)$. The endomorphism in Lemma 9.1 lies in the real cyclotomic field $K_N^+ = \mathbb{Q}(\zeta_N + \zeta_N^{-1})$. Evaluating the logarithm identity on each primitive holomorphic eigenform gives

$$\sigma_k(q) = \frac{I_{N,k}(s)}{I_{N,k}(1)}, \qquad 1 \leq k < N/2, \quad \gcd(k,N) = 1. \tag{9.4}$$

All these values lie in $(0,1)$. If $k < \ell$, then

$$\begin{aligned} &I_{N,k}(s)I_{N,\ell}(1) - I_{N,\ell}(s)I_{N,k}(1) \\ &\quad = \int_0^s \int_s^1 (t^{k-1}r^{\ell-1} - t^{\ell-1}r^{k-1}) w_N(t) w_N(r)\, \mathrm{d}r\, \mathrm{d}t > 0. \end{aligned} \tag{9.5}$$

Consequently the conjugates in (9.4) are distinct and

$$\mathbb{Q}(q) = K_N^+, \qquad [\mathbb{Q}(q) : \mathbb{Q}] = \varphi(N)/2. \tag{9.6}$$

Thus an algebraic probability with a conjugate outside $(0,1)$, which is not totally real, or which has the wrong degree, cannot have an algebraic inverse value. These are necessary, not sufficient, conditions for an exception.

The exceptional set

$$\mathcal{E}_N = \{s \in \overline{\mathbb{Q}} \cap (0,1) : R_N(s) \in \overline{\mathbb{Q}}\}$$

is finite. Indeed, $Z_N = \pi([\mathcal{H}_N - O])$ is an irreducible image curve containing zero and generating A_N. Its image is not a point because $\pi^*\eta_1 \neq 0$. Since $\dim A_N \geq 2$, it cannot be a torsion translate of an elliptic subvariety. Manin–Mumford therefore gives finitely many torsion points on Z_N [12]. The map from the smooth projective curve to its image is finite, and (9.3) proves the claim. No effective bound on the number of exceptions is claimed by this step.

9.4 Two algebraic irrational ratios for every exponent

For $q = (\sqrt{5}-1)/2$ or $q = 1/\sqrt{2}$, a conjugate is negative. Thus (9.4) excludes an algebraic inverse when $\varphi(N) \geq 4$. For $N = 3, 4, 6$ the real endomorphism field is $\mathbb{Q}$, so Lemma 9.1 likewise excludes either irrational ratio. For $N = 2$ the sine expression is excluded by Gelfond–Schneider. Hence both fixed ratios have transcendental inverses for every $N \geq 2$; $N = 1$ is the explicit algebraic exception $2q - q^2$. For the golden-ratio division at $N = 8$,

$$\begin{aligned} s &= 0.71630271941414350984526808090819462724 24\ldots, \\ s^8 &= 0.06930631363018070542289306644560096724 87\ldots. \end{aligned}$$

The latter is the corresponding beta quantile.

10 Local abelian logarithms and primitive divisors

The torsion reformulation admits a different exclusion mechanism: exhibit a single nonzero local logarithm. Complex and ℓ-adic integral values are not identified. It is the algebraic assertion of torsion that passes between embeddings.

Lemma 10.1 (Local nonvanishing, allowing ramification)**.** *Let $N \geq 3$, let $\ell \geq 5$ be prime with $\ell \nmid N$, and let an algebraic $s \in (0,1)$ be a unit in a finite extension of $\mathbb{Q}_\ell$. Normalize $v_\ell(\ell) = 1$. If*

$$h = v_\ell(1 - s^N) > \frac{2}{\ell - 1}, \tag{10.1}$$

then $\pi([P(s) - O])$ is not torsion. In particular $R_N(s)$ is transcendental.

Proof. Hensel lifting gives $r^N = 1$ with $r \equiv s$ in the residue field. At $W_r = (r, 0)$ use y as a parameter, and set $z = \sqrt{1 - s^N}$. Then $v_\ell(z) = h/2 > 1/(\ell - 1)$ and

$$x = r(1 - y^2)^{1/N}, \qquad \frac{dx}{y} = -\frac{2r}{N}(1 - y^2)^{1/N - 1}\,\mathrm{d}y.$$

The curve has good reduction since $\ell \nmid 2N$. In this small formal neighborhood the abelian integral is its logarithm coordinate

$$\mathcal{L} = -\frac{2r}{N} \sum_{j \geq 0} \frac{(1 - 1/N)_j}{j!(2j + 1)} z^{2j+1}. \tag{10.2}$$

The functorial logarithm and its Newton–Leibnitz property are as in [13]; the displayed valuation places the formal group coordinates inside the small logarithm convergence domain.

The coefficients $(1 - 1/N)_j/j!$ belong to $\mathbb{Z}_\ell$. If $k = v_\ell(2j + 1)$, then $1 + k(\ell - 1) \leq \ell^k \leq 2j + 1$, whence

$$v_\ell(2j + 1) \leq \frac{2j}{\ell - 1}.$$

The valuation of each term with $j \geq 1$, minus that of the first term, is at least

$$jh - v_\ell(2j + 1) \geq j\left(h - \frac{2}{\ell - 1}\right) > 0,$$

and this lower bound tends to infinity. The first term is uniquely minimal, so $v_\ell(\mathcal{L}) = h/2$ and $\mathcal{L} \neq 0$.

If $\pi([P(s) - O])$ were torsion, then so would $\pi([P(s) - W_r])$, by (9.2). Functoriality, torsion annihilation, and $\pi^*\eta_1 = dx/y$ would give

$$\mathcal{L} = \eta_1\big(\log_{A_N} \pi([P(s) - W_r])\big) = 0,$$

a contradiction. Now use (9.3). □

The strict inequality in (10.1) is part of the statement. If s lies in an unramified extension and $v_\ell(1 - s^N) \geq 1$, it holds automatically. This recovers the original unramified local test.

10.1 All rational locations, without an exponent exception

Theorem 10.2. *For every $N \geq 3$ and every $s \in \mathbb{Q} \cap (0,1)$, $R_N(s)$ is transcendental.*

Proof. Write $s = a/b$, where $0 < a < b$ and $\gcd(a, b) = 1$. Bang–Zsigmondy provides a primitive prime divisor ℓ of $b^N - a^N$, except at $(b, a, N) = (2, 1, 6)$; this precise form is recalled in [15]. Primitivity makes the residue of b/a have order N, so $N \mid \ell - 1$. Hence $\ell \geq 5$, $\ell \nmid 2Nab$, and $v_\ell(1 - s^N) \geq 1$. Apply Lemma 10.1. At the sole exception, $2^6 - 1 = 63$ and $\ell = 7$ still satisfies the local conditions. A primitive divisor is a sufficient device for choosing the prime, not a necessary hypothesis of the local lemma. □

For example, $R_{12}(1/2)$ can use $\ell = 13$ because $2^{12} - 1 = 3^2 \cdot 5 \cdot 7 \cdot 13$. Its value is

$$0.450017109099808036111667642573989081152 6\ldots.$$

Other examples are $R_{18}(1/2)$, $R_{24}(1/2)$, and $R_{12}(2/3)$, using primes 19, 241, and 61, respectively. The geometric meaning is fractional arclength at a specified rational normalized radius, not a rational fraction of the arclength specified in advance.

Corollary 10.3. *If $N = 2^h M$ with $M \geq 3$, and $s^{2^h} \in \mathbb{Q} \cap (0,1)$, then $R_N(s)$ is transcendental.*

Proof. Write $s^{2^h} = a/b$ and choose the good prime from the proof for M. The polynomial $X^{2^h} - a/b$ has simple roots modulo this odd prime, and all its roots lie in unramified extensions. Also $1 - s^N = 1 - (a/b)^M$ has positive integral valuation. Apply Lemma 10.1. The $M = 6, a/b = 1/2$ exception again uses 7. □

For instance $R_{12}(2^{-1/4}) = 0.760658039134571678695270791418183917436 7\ldots$ is transcendental and equals the regularized incomplete beta value $\mathrm{I}_{1/8}(1/12, 1/2)$.

11 Uniformity for bounded algebraic degree

The preceding local lemma remains valid in ramified fields. Combining it with a uniform primitive-ideal theorem gives a height-independent result.

External theorem 11.1 (Voutier)**.** *Let A, B be nonzero coprime algebraic integers in a number field, suppose A/B is not a root of unity, and put $d = [\mathbb{Q}(A/B) : \mathbb{Q}]$. If*

$$N > \max\{2(2^d - 1),\ 4000(d\log(3d))^{12}\},$$

then $B^N - A^N$ has a prime-ideal divisor which divides no earlier $B^j - A^j$, $1 \leq j < N$.

This is [17, Theorem 2(i)], not the Lucas–Lehmer-only part of that theorem. The degree in the bound is the degree of the ratio, rather than the degree of an arbitrary ambient number field.

Theorem 11.2. *For an integer $D \geq 1$, define*

$$C(D) = \max\{3^D,\ 4000(D\log(3D))^{12}\}. \tag{11.1}$$

If algebraic $s \in (0,1)$ has degree at most D and $N > C(D)$, then $R_N(s)$ is transcendental. The bound is independent of height, denominators, and the discriminant of $\mathbb{Q}(s)$.

Proof. Let $K = \mathbb{Q}(s)$, and write the fractional ideal $(s) = \mathfrak{a}\mathfrak{b}^{-1}$ with coprime integral ideals. Some power $\mathfrak{b}^h = (b_0)$ is principal. In a finite extension containing $B^h = b_0$, one has $(B) = \mathfrak{b}O_L$, because equality of hth powers of fractional ideals implies equality. Set $A = sB$. Then A, B are coprime algebraic integers, while their ratio still has degree $[K : \mathbb{Q}]$. Apply External theorem 11.1 in L.

Contract a primitive prime ideal to a prime $\mathfrak{p}$ of K. The injection of residue fields preserves the exact order N of s. If the residue characteristic is ℓ and the residue degree in K is f, then

$$N \mid \ell^f - 1, \quad \ell \nmid N, \quad v_{\mathfrak{p}}(1 - s^N) \geq 1.$$

Let e be its ramification index in the original field K. Then $ef \leq [K : \mathbb{Q}] \leq D$ and $v_\ell(1 - s^N) \geq 1/e$. If $\ell \leq 2e + 1$, then

$$N + 1 \leq \ell^f \leq (2e + 1)^f \leq 3^{ef} \leq 3^D,$$

contradicting $N > C(D)$. Thus $\ell > 2e + 1$, so $1/e > 2/(\ell - 1)$ and $\ell \geq 5$. Apply Lemma 10.1. The comparison uses e, f in K, not in the potentially much larger principalizing extension. □

For $D = 2$, $C(2) < 20\,000\,000\,000$. This is explicit but very large; no optimality is claimed. For each fixed algebraic s, only finitely many exponents can give an algebraic ratio. More uniformly, an exceptional sequence with exponents tending to infinity, if one exists, must have input degrees tending to infinity.

11.1 A small-exponent ramified quadratic example

Let $s_0 = 3 - \sqrt{7} \in (0, 1)$. No positive power of s_0 is rational, since its conjugate $3 + \sqrt{7} > 1$ has a different positive power. Thus it is not covered by the power-root corollary above. In $\mathbb{Q}_7(\sqrt{7})$, $v_7(\sqrt{7}) = 1/2$. If $N = 6m$ with $7 \nmid m$, then $3^N \equiv 1 \pmod 7$ and expansion gives

$$1 - (3 - \sqrt{7})^N = 1 - 3^N + N3^{N-1}\sqrt{7} + \sum_{j=2}^{N} (-1)^{j+1} \binom{N}{j} 3^{N-j} (\sqrt{7})^j.$$

The linear term has valuation $1/2$, whereas every other term has valuation at least one. Hence $v_7(1 - s_0^N) = 1/2 > 1/3$, and

$$R_{6m}(3 - \sqrt{7}) \notin \overline{\mathbb{Q}} \qquad (m \geq 1,\ 7 \nmid m). \tag{11.2}$$

In particular

$$1 - (3 - \sqrt{7})^{12} = -524361663 + 198190080\sqrt{7},$$

and

$$R_{12}(3 - \sqrt{7}) = 0.31883299555500762110964032542302162981 65\ldots.$$

Here the local logarithm has valuation $1/4$. This small-exponent result does not use the large constant (11.1).

12 Fermat beta integrals, polynomial weights, and second-kind forms

The preceding sections exploit a particular hyperelliptic character. We now change the curve rather than merely enlarge an exponent. A complete torsion classification for Fermat quotients makes it possible to treat all algebraic interior inputs for another family. The polynomial-weight argument then requires a stronger statement than transcendence of a ratio: it requires linear independence with an algebraic constant term.

Theorem 12.1 (Beta independence and weighted stability)**.** *Let p, r be distinct primes with $\max(p, r) \geq 5$, and put*

$$a = i/p, \quad b = j/r, \qquad 1 \leq i < p, \quad 1 \leq j < r.$$

No restriction $a + b < 1$ is imposed. For every $x \in \overline{\mathbb{Q}} \cap (0, 1)$,

$$1, \quad \mathrm{B}(a, b), \quad \mathrm{B}_x(a, b) \quad \text{are linearly independent over } \overline{\mathbb{Q}}. \tag{12.1}$$

Let $W(t) = \sum w_k t^k \in \overline{\mathbb{Q}}[t]$, and define

$$c_W = \sum_k w_k \frac{(a)_k}{(a + b)_k}. \tag{12.2}$$

Then, at every such algebraic x,

$$\int_0^x W(t) t^{a-1} (1 - t)^{b-1}\, \mathrm{d}t \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad c_W = 0. \tag{12.3}$$

If $c_W \neq 0$, the integral normalized by its value at 1 is transcendental. In particular, if W has real algebraic coefficients and is strictly positive on $(0, 1)$, every interior algebraic-probability quantile of its normalized density is transcendental.

All nonnegative integral parameter shifts are included by taking $W = t^n(1-t)^m$. There is no bound on the degree of W or on the degree or height of x. Equal prime denominators, general composite denominators, and the prime pair $\{2, 3\}$ are not included.

12.1 The curve and the pole that cannot be ignored

Work on the smooth projective model of

$$C: \quad U^p + V^r = 1. \tag{12.4}$$

It has a unique point at infinity and genus $(p-1)(r-1)/2$. The differential

$$\eta = U^{i-1}V^{j-r}\,\mathrm{d}U \tag{12.5}$$

is regular at every finite point. At $V = 0$, differentiation of the curve equation supplies the factor V^{r-1} needed for regularity. At infinity,

$$\operatorname{ord}_\infty \eta = pr - ri - pj - 1. \tag{12.6}$$

Thus $a + b < 1$ gives a holomorphic form. If $a + b > 1$, the form has a pole of order at least two, with residue zero. Equality $a + b = 1$ is impossible for the specified prime denominators. The vanishing residue also follows from the global residue theorem, since infinity is the only possible pole. The real path

$$P_x = (x^{1/p}, (1-x)^{1/r}), \qquad 0 \le x \le 1,$$

satisfies

$$P^*\eta = \frac{1}{p}x^{a-1}(1-x)^{b-1}\,\mathrm{d}x. \tag{12.7}$$

For example $a = 2/3$, $b = 2/5$ gives $C : U^3 + V^5 = 1$ and $\eta = UV^{-3}\,\mathrm{d}U$, with $\operatorname{ord}_\infty \eta = -2$. Treating it as a holomorphic differential on the ordinary Jacobian would be incorrect.

12.2 Generalized Jacobians retain the polar part

Choose $M \ge 2$ at least as large as the pole order. The generalized Jacobian for the modulus $M\infty$ fits into

$$0 \longrightarrow \mathbb{G}_a^{M-1} \longrightarrow G \xrightarrow{\varrho} J \longrightarrow 0, \qquad J = \operatorname{Jac}(C). \tag{12.8}$$

There is no torus because the modulus has a single support point. This is the standard modulus construction [18]. With $O = (0, 1)$ as base point, an algebraic Abel map $j_O : C \setminus \{\infty\} \to G$ and an algebraic invariant differential $\widetilde{\eta}$ satisfy

$$j_O^*\widetilde{\eta} = \eta. \tag{12.9}$$

One can check the differential condition locally rather than infer it solely from an analytic formula. The pinched local ring is $\overline{\mathbb{Q}} + t^M\overline{\mathbb{Q}}[[t]]$. The Rosenlicht condition asks that $\operatorname{Res}(f\eta) = 0$ for every element of this ring. Constants are handled by the zero residue, and multiplication by t^M removes the pole. The corresponding dualizing differential therefore gives an invariant differential of the generalized Jacobian. See [19] for the residue formulation and integral description of the Abel map. All the modulus and Picard constructions used here are over $\overline{\mathbb{Q}}$.

The vector-group fiber in (12.8) is contractible as a complex manifold. Consequently

$$H_1(G(\mathbb{C}), \mathbb{Q}) \simeq H_1(J(\mathbb{C}), \mathbb{Q}). \tag{12.10}$$

The rotation $T(U,V) = (\zeta_p U, \zeta_r V)$ preserves the modulus and acts on G. Its action on the Abel map can differ by translation because it need not fix O; invariant differentials are unaffected by that translation. On $H_1(J,\mathbb{Q})$ its eigenvalues are exactly $\zeta_p^u \zeta_r^v$, $1 \le u < p$, $1 \le v < r$, each once. Therefore

$$\chi_{T_*} = \Phi_{pr}, \qquad H_1(J,\mathbb{Q}) \simeq K = \mathbb{Q}(\zeta_{pr}) \tag{12.11}$$

as a rank-one K-module. This also follows from the homology calculation in [11, Proposition 2.4.4 and Corollary 2.4.5]. Unlike a general composite hyperelliptic exponent, the entire Jacobian is the relevant rational character module.

12.3 A nonzero period for a possibly second-kind form

It is not enough here to say that the differential is nonzero: a nonzero exact differential can have every closed period zero. We exhibit a specific nonzero period. Let ℓ be the real path from $O = (0,1)$ to $W = (1,0)$, and set

$$S(U,V) = (\zeta_p U, V), \qquad R(U,V) = (U, \zeta_r V).$$

The map S fixes O, and R fixes both W and SW. Hence

$$\Gamma = (1 - R_*)(1 - S_*)\ell$$

is a closed integral cycle avoiding infinity. Directly,

$$\int_\Gamma \eta = (1-\zeta_r^j)(1-\zeta_p^i)\frac{\mathrm{B}(a,b)}{p} \neq 0. \tag{12.12}$$

The Abel map realizes this as a period on G. If $\sigma : K \hookrightarrow \mathbb{C}$ sends T to $\xi = \zeta_p^i \zeta_r^j$, the eigencharacter relation and (12.11) now imply

$$\ker \mathcal{P}_{\widetilde{\eta}} = 0, \qquad \operatorname{im} \mathcal{P}_{\widetilde{\eta}} = \Omega\sigma(K), \quad \Omega \neq 0. \tag{12.13}$$

As before, this is injectivity on rational homology, not on its complex scalar extension.

12.4 Projection rigidity, not vanishing of the entire logarithm

The additive directions in G require a modification of Lemma 9.1. A logarithm can have algebraic coordinates in those directions without being zero.

Lemma 12.2 (Generalized projection rigidity)**.** *If $w \in \mathrm{Lie}(G)_{\mathbb{C}}$ exponentiates to an algebraic point and $\widetilde{\eta}(w) \in \overline{\mathbb{Q}}$, then $\mathrm{d}\varrho(w) = 0$. If u, v exponentiate to algebraic points, $\bar{v} = \mathrm{d}\varrho(v) \neq 0$, and*

$$\widetilde{\eta}(u) - \lambda\widetilde{\eta}(v) = c, \qquad \lambda, c \in \overline{\mathbb{Q}}, \tag{12.14}$$

then $\lambda \in \sigma(K)$ and

$$\mathrm{d}\varrho(u) = a_* \bar{v} \tag{12.15}$$

for the corresponding $a \in K \subseteq \mathrm{End}^0(J)$.

Proof. For the first assertion, let $c = \widetilde{\eta}(w)$ and apply External theorem 2.3 to $(w,1)$ on $G \times \mathbb{G}_a$ and the hyperplane $\widetilde{\eta}(Z) - cT_0 = 0$. Let H be the resulting connected subgroup. If its image $B \subseteq J$ were positive dimensional, the kernel of $H \to B$ would be a vector group. Hence rational homology maps onto $H_1(B,\mathbb{Q})$, and a cycle with nonzero image in J could be chosen. The integral of the additive coordinate differential around this cycle is zero. The hyperplane then forces its $\widetilde{\eta}$-period to vanish, contrary to (12.13). Thus $B = 0$.

For the second assertion, use $G \times G \times \mathbb{G}_a$, the logarithm $(u, v, 1)$, and (12.14). The second abelian projection is positive dimensional since $\bar{v} \neq 0$. Choose a rational cycle whose second abelian projection is nonzero. Integrating the hyperplane relation gives λ as a ratio of two periods in $\Omega\sigma(K)$, with nonzero denominator. Thus $\lambda \in \sigma(K)$.

It is not necessary to embed the field K into the entire endomorphism algebra of G. Choose $A(X) \in \mathbb{Q}[X]$ with $A(\xi) = \lambda$ and use the actual quasi-endomorphism $A(T)$ of G. It induces the required a on J. After clearing denominators, $mu - (mA(T))_* v$ exponentiates to an algebraic point and has algebraic $\widetilde{\eta}$-coordinate mc. The first assertion gives (12.15). □

If $\bar{v}$ is a torsion logarithm, it is a rational period, and (12.15) makes $\mathrm{d}\varrho(u)$ a rational period as well. Only the image in the ordinary Jacobian is forced to be torsion; no such claim is made about the point of G itself.

12.5 Torsion packets force three-term independence

The divisors

$$\operatorname{div}(V - 1) = p(O - \infty), \qquad \operatorname{div}(U - 1) = r(W - \infty) \tag{12.16}$$

show that $[W - O]$ is torsion in J. Choose actual logarithms $u_x, v = u_1$ along the real path. Then

$$\widetilde{\eta}(u_x) = \mathrm{B}_x(a, b)/p, \qquad \widetilde{\eta}(v) = \mathrm{B}(a, b)/p. \tag{12.17}$$

The projected logarithm $\bar{v}$ is a nonzero rational period. Its nonvanishing is independently witnessed by the positive integral of the holomorphic differential with indices $i = j = 1$, since $1/p + 1/r < 1$.

Suppose $c_0 + c_1\mathrm{B}(a, b) + c_2\mathrm{B}_x(a, b) = 0$ with algebraic coefficients. If $c_2 \neq 0$, Lemma 12.2 makes $[P_x - O]$ torsion in J, and hence also $[P_x - \infty]$ is torsion. For odd p, r, the transformation $(X, Y) = (-U, V)$ gives $Y^r = X^p + 1$. Arul's complete classification [11, Theorem 6.2.21] excludes noncuspidal torsion points for these distinct odd prime exponents. Our point has both coordinates nonzero, so it is excluded.

If one prime is 2, interchange variables if necessary and write $r = 2$, $p \geq 5$. The real image has $X = -x^{1/p} \in (-1, 0)$. For $p \geq 7$ there is no additional torsion orbit. For $p = 5$ the additional orbit satisfies $X^5 = 4$, and therefore misses this interval. These are precisely the exceptional cases retained from the classification, not ignored by it. Thus $c_2 = 0$. If $c_1 \neq 0$, the beta period would be algebraic, and the first part of Lemma 12.2 applied to v would contradict $\bar{v} \neq 0$. Hence all three coefficients vanish, proving (12.1).

In the holomorphic subcase the same reasoning can be performed directly on J: Lemma 9.1 supplies the torsion reduction and Lemma 2.4 removes the algebraic constant term. The generalized argument is what permits $a + b > 1$.

12.6 Exact polynomial reduction and its algebraic exception

Set $D(x) = x^a(1 - x)^b$. Standard incomplete-beta recurrences [23] give a polynomial $P_W \in \overline{\mathbb{Q}}[x]$ such that

$$\int_0^x W(t)t^{a-1}(1 - t)^{b-1}\,\mathrm{d}t = c_W\mathrm{B}_x(a, b) + D(x)P_W(x). \tag{12.18}$$

For clarity, a constructive recurrence for $W = t^k$ is

$$c_0 = 1, \qquad P_0 = 0, \; c_{k+1} \quad = \frac{a + k}{a + b + k}c_k, P_{k+1} \quad = \frac{a + k}{a + b + k}P_k - \frac{x^k}{a + b + k}. \tag{12.19}$$

Linear combinations handle every polynomial. Differentiation verifies

$$x(1-x)P'_W + [a-(a+b)x]P_W = W - c_W. \tag{12.20}$$

The endpoint term vanishes at both 0 and 1, so the total integral is $c_W \mathrm{B}(a,b)$. At algebraic x, $D(x)P_W(x)$ is algebraic. Thus (12.1) proves (12.3); it also excludes an algebraic normalized value when $c_W \neq 0$. Positivity gives $c_W > 0$ and strict monotonicity, which supplies the quantile statement. This completes the proof of Theorem 12.1.

For a fixed degree bound on W, the algebraic cases form exactly the linear hyperplane $c_W = 0$. For example, with $a = 2/3, b = 2/5$,

$$\int_0^x (8t-5)t^{-1/3}(1-t)^{-3/5}\,\mathrm{d}t = -\frac{15}{2}x^{2/3}(1-x)^{2/5}. \tag{12.21}$$

The weight changes sign and is not a probability density. The example prevents an incorrect claim that every nonzero weighted integral is transcendental.

12.7 The exact linear dimension of all weighted values

For fixed algebraic $x \in (0,1)$ and admissible positive W, write

$$F_0(x) = \mathrm{B}_x(a,b)/\mathrm{B}(a,b), \qquad F_W(x) = F_0(x) + \frac{h_W(x)}{\mathrm{B}(a,b)}, \quad h_W(x) \in \overline{\mathbb{Q}}.$$

Consequently

$$\dim_{\overline{\mathbb{Q}}} \operatorname{span}\big(\{1\} \cup \{F_W(x) : W > 0\}\big) = 3. \tag{12.22}$$

The upper bound follows from the three displayed directions. Their independence follows from (12.1) after multiplication by $\mathrm{B}(a,b)$. Taking $W = 1, t$ realizes the third direction, since

$$F_t(x) = F_0(x) - \frac{D(x)}{a\mathrm{B}(a,b)}.$$

Equation (12.22) concerns linear dimension, not transcendence degree.

12.8 Explicit first-kind and second-kind examples

For $a = 1/3, b = 1/5$, let $B_0 = \mathrm{B}(a,b)$ and $D = x^{1/3}(1-x)^{1/5}$. If G denotes the $\mathrm{Beta}(4/3, 6/5)$ distribution function, and H the distribution obtained with weight $1+t^2$, then

$$B_0 G(x) = \mathrm{B}_x(1/3,1/5) + (8x-3)D(x), \; B_0 H(x) \quad = \mathrm{B}_x(1/3,1/5) - \frac{15}{71}(5+2x)D(x). \tag{12.23}$$

Here $c_{t(1-t)} = 15/184$ and $c_{1+t^2} = 71/46$. In particular

$$71H(1/2) - 161F_0(1/2) + 90G(1/2) = 0,$$

whereas $1, F_0(1/2), G(1/2)$ are linearly independent. The median of G is

$$0.534154161604607302042768822 6978928570926\ldots,$$

and is transcendental by Theorem 12.1.

For the second-kind example $a = 2/3, b = 2/5$, set $B_0 = \mathrm{B}(2/3, 2/5)$ and $D = x^{2/3}(1-x)^{2/5}$. The distribution function $G = \mathrm{I}_x(5/3, 7/5)$ satisfies

$$B_0 G(x) = \mathrm{B}_x(2/3, 2/5) + (4x - 3/2)D(x), \tag{12.24}$$

with $c_{t(1-t)} = 15/124$. Its median is

$$m = 0.55398928402851700561828807433349865293 70\ldots. \tag{12.25}$$

If m were algebraic, (12.24) at $G(m) = 1/2$ would contradict (12.1). This is not covered by the integral-parameter-sum HPD theorem: $5/3 + 7/5 = 46/15$.

The hypotheses remain material. Equal denominators allow symmetric laws, for example $\mathrm{Beta}(1/3, 1/3)$ with median $1/2$. The pair $\{2, 3\}$ leads to an elliptic curve with noncuspidal torsion points and algebraic rational divisions. No classification for arbitrary rational-function weights or general composite denominators is asserted. Dhillon–Saha [22] is related work on incomplete-beta values; a complete comparison with every theorem of that paper is not part of this consolidation and no priority conclusion is drawn from its abstract.

13 Attempted transfer to the Euler–Mascheroni constant

This section records an attempted transfer, not a proof of the irrationality or transcendence of γ. It also does not certify every preceding manuscript argument by virtue of having compared it with the gamma problem. The purpose is to determine precisely which hypotheses of the successful mechanisms survive and which additional arithmetic implication is missing.

The target is the constant in (1.1), rather than the function $\Gamma(z)$ as an element of a differential field. For example, $\Gamma(n) = (n-1)!$ is algebraic at every positive integer, despite the transcendental nature of the function. Likewise, transcendence of some higher derivatives does not identify the arithmetic of $\Gamma'(1)$.

13.1 A normalization ambiguity invisible to beta quotients

The standard connection is

$$\mathrm{B}(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}. \tag{13.1}$$

For an arbitrary $c \in \mathbb{C}$, put $\Gamma_c(z) = e^{cz}\Gamma(z)$. Then

$$\frac{\Gamma_c(a)\Gamma_c(b)}{\Gamma_c(a+b)} = \mathrm{B}(a, b), \qquad \psi_c(z) = c + \psi(z), \quad \psi_c(1) = c - \gamma. \tag{13.2}$$

Thus the beta quotient alone cannot detect the linear term in the logarithm of the gamma function. In particular,

$$\partial_a \log \mathrm{B}(a, b) = \psi(a) - \psi(a+b)$$

cancels the common constant $-\gamma$. This is an information limitation of a beta-quotient-only transfer, not a claim that Γ_c satisfies every normalization or recurrence of Γ. In general it does not. Adding normalization or growth data changes the problem; (13.2) does not rule out every possible argument involving beta integrals.

13.2 Confluence returns gamma, but not an arithmetic lower bound

For $\Re z > 0$, define

$$G_M(z) = M^z \mathrm{B}(z, M+1) = \int_0^M t^{z-1}(1 - t/M)^M \,\mathrm{d}t = \frac{M^z M!}{z(z+1)\cdots(z+M)}. \tag{13.3}$$

Dominated convergence gives $G_M(z) \to \Gamma(z)$. The Laurent expansion at zero is exact to first order:

$$G_M(z) = \frac{1}{z} + \log M - H_M + O(z). \tag{13.4}$$

Its finite part tends to $-\gamma$. This is a valid analytic connection, but it returns to the classical harmonic-number approximation. Although $H_M - \log M$ is transcendental for $M > 1$, that fact alone does not classify its limit. The same warning applies to the regularized Ein limit in (3.11): its explicit appearance of γ and a dilogarithm is an identity, not an automatic transcendence transfer.

13.3 Polynomial logarithmic moments lock the gamma and tail coefficients

The polynomial-weight method can be tested directly. Let $P(t) = \sum_{k=0}^{d} p_k t^k \in \overline{\mathbb{Q}}[t]$ and set

$$B_P = \sum_{k=0}^{d} p_k k!, \qquad A_P = \sum_{k=0}^{d} p_k k! H_k. \tag{13.5}$$

Differentiating the gamma recurrence gives $\Gamma'(k+1) = k!(H_k - \gamma)$ [23]. Therefore

$$J_P := \int_0^\infty P(t) e^{-t} \log t \, \mathrm{d}t = A_P - B_P \gamma. \tag{13.6}$$

These integrals converge absolutely. If $B_P = 0$, J_P is algebraic. If $B_P \neq 0$, then

$$J_P \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \gamma \in \overline{\mathbb{Q}}. \tag{13.7}$$

In particular, increasing the complexity of a polynomial does not by itself produce an independently established transcendence result for such a moment. For a positive real polynomial weight, $B_P = \int_0^\infty P(t) e^{-t} \, \mathrm{d}t > 0$, so the question remains exactly as strong as the gamma question.

To identify the tail, define the polynomials

$$Q_P(t) = \sum_{j=0}^{d} P^{(j)}(t), \qquad R_P(t) = \frac{Q_P(t) - Q_P(0)}{t}, \qquad C_P = \sum_{j \geq 0} R_P^{(j)}(1). \tag{13.8}$$

The last sum is finite; it is zero when $R_P = 0$. Directly,

$$Q_P - Q_P' = P, \qquad Q_P(0) = B_P, \qquad \int_0^\infty e^{-t} R_P(t) \, \mathrm{d}t = A_P.$$

The last identity follows first for monomials and then by linearity. Integration by parts, using $\log 1 = 0$, gives

$$\begin{aligned} T_P := \int_1^\infty P(t) e^{-t} \log t \, \mathrm{d}t &= \int_1^\infty e^{-t} \frac{Q_P(t)}{t} \, \mathrm{d}t \\ &= B_P E_1(1) + \frac{C_P}{e}. \end{aligned} \tag{13.9}$$

With $\eta = \mathrm{Ein}(1) = \gamma + E_1(1)$, the lower piece is

$$L_P := -\int_0^1 P(t) e^{-t} \log t \, \mathrm{d}t = B_P \eta - A_P + \frac{C_P}{e}. \tag{13.10}$$

The signs agree with $J_P = T_P - L_P$.

$P(t)$	A_P	B_P	C_P	J_P
1	0	1	0	$-\gamma$
t	1	1	1	$1-\gamma$
t^2	3	2	4	$3-2\gamma$
t^3	11	6	17	$11-6\gamma$
$t-1$	1	0	1	1

The same coefficient B_P multiplies $-\gamma$ in (13.6) and $E_1(1)$ in (13.9). Consequently any finite linear combination cancelling the displayed $E_1(1)$ coefficient also cancels the gamma coefficient. This statement is at the level of the explicit coefficients; it does not assume an unproved algebraic relation or independence between $E_1(1)$ and e. An exact counterexample to indiscriminate transcendence claims is

$$\int_0^\infty (t-1)e^{-t}\log t\,\mathrm{d}t = 1. \tag{13.11}$$

The scope is fixed e^{-t}, one factor $\log t$, finite polynomial weights, and linear elimination. Rational-function weights, nonlinear combinations, several logarithms, and new height estimates are not excluded by this calculation.

13.4 Why a finite modulus does not directly accommodate the exponential

In Section 12, the pole order was finite and a modulus $M\infty$ produced a finite-dimensional generalized Jacobian. In

$$-\gamma = \int_0^\infty e^{-t}\log t\,\mathrm{d}t,$$

the substitution $u = 1/t$ introduces $e^{-1/u}$, an essential singularity. A finite algebraic covering does not turn that exponential essential singularity into a finite-order pole. Thus the current expression is not made into an algebraic meromorphic differential by the earlier modulus construction. This is not a proof that γ has no other period representation. It identifies a failure of this particular direct construction. Moreover, assuming γ algebraic does not automatically supply an algebraic point whose logarithm is the displayed integral, nor a torsion point on the Fermat curves. Those were essential hypotheses of the preceding analytic-subgroup and local arguments.

13.5 Two arithmetic blocks containing the target exactly

For positive real z,

$$\mathrm{Ein}(z) = \gamma + \log z + E_1(z). \tag{13.12}$$

Define the entire E-function

$$A(z) = e^z\,\mathrm{Ein}(z) = \sum_{n\ge 0}\frac{H_n}{n!}z^n \tag{13.13}$$

and the formal factorial series at infinity

$$\widehat{D}(z) = \sum_{n\ge 0}(-1)^n n!\,z^{-n-1}. \tag{13.14}$$

The latter diverges for every fixed nonzero complex z, but its positive-direction Borel–Laplace sum is

$$D_0(z) = \int_0^\infty \frac{e^{-t}}{z+t}\,\mathrm{d}t = e^z E_1(z). \tag{13.15}$$

For each algebraic candidate r, form

$$\widehat{\Psi}_r(z) = A(z) - re^z - \widehat{D}(z). \tag{13.16}$$

Its defining coefficients use harmonic numbers, factorials, and r; they do not require a decimal approximation to γ. Positive-direction summation gives the exact formula

$$\Psi_{r,0}(z) = e^z(\gamma - r + \log z), \qquad \Psi_{r,0}(1) = e(\gamma - r). \tag{13.17}$$

The arithmetic framework for such mixed objects is that of Fischler–Rivoal [20]. The formal mixed class must be distinguished from an arbitrary analytic solution of a differential equation.

13.5.1 A fixed differential equation is not a value theorem

The identities

$$A' = A + (e^z - 1)/z, \qquad D_0' = D_0 - 1/z$$

give $\Psi_{r,0}' = \Psi_{r,0} + e^z/z$. Eliminating e^z yields

$$zY'' + (1 - 2z)Y' + (z - 1)Y = 0. \tag{13.18}$$

Every algebraic candidate r gives a solution of the same equation, whose solution space is spanned by e^z and $e^z \log z$. The coefficient $\gamma - r$ is a connection constant specifying the arithmetic solution, not a coefficient of (13.18). Finding this equation therefore does not by itself exclude any candidate r. The unresolved issue concerns compatibility of the two arithmetic blocks, not existence of the analytic solution.

13.6 An arithmetic division obstruction and the missing implication

There is a concrete obstruction on one side of a potential zero theorem.

Proposition 13.1. *For algebraic r, there is no identity*

$$\widehat{\Psi}_r(z) = (z - 1)\left(F(z) + \sum_{n \geq 1} b_n z^{-n}\right) \tag{13.19}$$

with F an E-function and all b_n algebraic. In particular such a factorization does not exist in the mixed arithmetic class.

Proof. Comparison of the nonnegative powers gives

$$A(z) - re^z = (z - 1)F(z) + b_1.$$

Evaluating this entire-function identity at 1 gives $b_1 = e\,\mathrm{Ein}(1) - re$. This is incompatible with the algebraic independence of e and $\mathrm{Ein}(1)$, a subcase of Theorem 3.1 (or the classical E-function argument in [21]). □

The missing implication is

$$\Psi_{r,0}(1) = 0 \quad \Longrightarrow \quad \widehat{\Psi}_r = (z - 1) \times \text{a mixed arithmetic function.} \tag{13.20}$$

An analytic zero permits analytic division on that branch. It does not prove that the quotient has the required algebraic coefficients and arithmetic denominator growth. A general zero-division assertion of this kind is Conjecture 3 in [20], not an established input available for unconditional use.

In the present family, asserting (13.20) for every algebraic r is equivalent to transcendence of γ. One implication uses Proposition 13.1 and (13.17); the other is vacuous because if γ is transcendental the premise never holds for algebraic r. Thus this is a sharply formulated target, but not a claim to have reduced the problem to an independently easier theorem. The same obstruction had already appeared in the earlier exploratory record recalled during the research conversation; it is not counted as a newly discovered result in this consolidation.

There is also an instructive monodromy check. Under the hypothetical identification $r = \gamma$, the quotient on the positive branch is

$$\frac{e^z \log z}{z - 1},$$

which has a removable singularity at 1. Continuing once around 0 adds $2\pi i e^z/(z - 1)$, introducing a pole on the other branch. This separates branchwise removability from a global arithmetic regularity statement; it is not itself a transcendence proof.

13.7 The two direct local series have disjoint convergence domains

A local nonvanishing calculation succeeded in Section 10 because an algebraic torsion assertion could be tested in one small residue disk. The two most direct gamma series do not even converge at a common nonzero local input.

Proposition 13.2. *Fix a prime ℓ, an algebraic ℓ-adic embedding of r, and normalize $v_\ell(\ell) = 1$. The series*

$$A_r(z) = \sum_{n\geq 0} \frac{H_n - r}{n!} z^n, \qquad \widehat{D}(z) = \sum_{n\geq 0} (-1)^n n! z^{-n-1}$$

converge, respectively, exactly on the open regions

$$v_\ell(z) > \frac{1}{\ell - 1}, \qquad v_\ell(z) < \frac{1}{\ell - 1}. \tag{13.21}$$

Neither converges on the common boundary.

Proof. Legendre's formula is

$$v_\ell(n!) = \frac{n - s_\ell(n)}{\ell - 1},$$

where $s_\ell(n)$ is the digit sum in base ℓ. Also $v_\ell(H_n) \geq -\lfloor \log_\ell n \rfloor$, so the harmonic coefficient changes only a logarithmic correction to the factorial radius. For necessity, take $n = \ell^k$. The unique smallest-valuation term in H_{ℓ^k} is $1/\ell^k$, hence

$$v_\ell(H_{\ell^k}) = -k, \qquad v_\ell(H_{\ell^k} - r) = -k$$

for sufficiently large k. At the boundary $v_\ell(z) = 1/(\ell - 1)$, the corresponding A_r term has valuation $-k + 1/(\ell - 1)$, whereas the corresponding $\widehat{D}$ term has valuation $-2/(\ell - 1)$. Neither sequence of terms tends to zero. The same subsequence excludes the wrong side of each boundary, and the factorial estimates give convergence on the stated sides. □

This does not prohibit ℓ-adic analytic continuation, Frobenius structures, or an overconvergent construction. It rules out a direct comparison of these two defining series at the same nonzero local point. Moreover a local nonvanishing statement alone would still require a valid arithmetic bridge from the complex gamma hypothesis.

13.8 A genuine conditional conclusion, kept conditional

Let $\mathcal{E}$ be the ring of algebraic-point values of E-functions, and let $\mathcal{D}$ be the ring of suitably summed arithmetic Gevrey series of order +1, in the precise framework of [20]. The following is an existing conjecture.

Conjecture 13.3 (Fischler–Rivoal)**.**

$$\mathcal{E} \cap \mathcal{D} = \overline{\mathbb{Q}}. \tag{13.22}$$

It implies the transcendence of γ. Indeed, put

$$\delta = eE_1(1) \in \mathcal{D}, \qquad \eta = \mathrm{Ein}(1), \qquad e, \eta \in \mathcal{E}.$$

The identity

$$\delta = e\eta - e\gamma \tag{13.23}$$

shows that an algebraic γ would put δ in $\mathcal{E}$ as well. Under Conjecture 13.3, δ would then be algebraic, contradicting the algebraic independence of e and η. This is a specialization of the more general conditional gamma-derivative result in [20, Theorem 2], not a proof of the conjecture.

An unconditional but weaker consequence is already immediate from the same algebraic independence: γ and δ cannot both be algebraic. This known alternative does not determine which of them is transcendental [21]. Similarly,

$$\Gamma''(1) = \gamma^2 + \frac{\pi^2}{6}$$

is entirely compatible with algebraic γ and transcendental $\Gamma''(1)$. Even an infinitude of transcendental higher derivatives would not by itself identify the first derivative.

13.9 What survives the attempted transfer

Mechanism examined	Outcome and exact limitation
Beta quotient and parameter differentiation	The common linear term of $\log\Gamma$ is invisible to the quotient; normalization information must be added.
Confluent beta limit	The finite part returns $H_M - \log M$; no new arithmetic lower bound for its limit has been supplied.
Polynomial one-log moments	The same coefficient controls γ and the displayed exponential-integral tail; finite linear cancellation removes both.
Finite-modulus generalized Jacobian	The exponential essential singularity is not a finite-order algebraic pole in the given representation.
Mixed E/factorial construction	A fixed second-order equation contains $e(\gamma - r)$ as the value of a selected arithmetic solution.
Arithmetic division	Formal division is excluded, but promotion of a summed zero to arithmetic divisibility remains unproved.
Direct local evaluation	The two natural series have disjoint convergence regions, including failure at the boundary.
Intersection conjecture	It would prove transcendence of γ, but is not established here.
Arithmetic status of γ	Neither irrationality nor transcendence is proved by this manuscript.

The missing work is an arithmetic noncancellation theorem for the selected mixed solution, or a sufficiently strong quantitative substitute. Repeating an unproved lifting assertion under a different name is not progress toward that theorem. Nor does abstract nonsplitting of an extension force a particular entry of an arbitrary comparison matrix to be transcendental. These distinctions are essential when evaluating a proposed transfer rather than merely producing another exact identity.

14 Numerical certification, reproducibility, and limitations

Three logically distinct layers occur in this work: an external theorem, a symbolic argument checking its hypotheses, and a finite computation. A successful numerical test does not prove the external theorem or turn a flawed hypothesis check into a valid proof. Conversely, an exact root interval can be certified without assuming that its center is an accurate floating-point approximation.

14.1 Positive series and explicit remainder estimates

For $0 < a, b < 1$ and $0 < x \le 1/2$,

$$\mathrm{B}_x(a,b) = x^a \sum_{j\ge 0} \frac{(1-b)_j}{j!(a+j)} x^j. \tag{14.1}$$

Every term is positive and the ratio of consecutive terms is less than x. If T is the first omitted term, the remainder is at most $T/(1-x)$. The complete integral is evaluated using

$$\mathrm{B}(a,b) = \mathrm{B}_{1/2}(a,b) + \mathrm{B}_{1/2}(b,a),$$

and values with $x > 1/2$ are obtained by subtraction from the complete integral. Algebraic powers are enclosed by integer-root computations on a rational grid. Polynomial corrections such as (12.18) are then evaluated with rational interval arithmetic, retaining the rounding error at every step.

Similarly,

$$I_{N,k}(s) = \int_0^s \frac{t^{k-1}\,\mathrm{d}t}{\sqrt{1-t^N}} = \sum_{j\ge 0} \frac{\binom{2j}{j} s^{Nj+k}}{4^j (Nj+k)} \tag{14.2}$$

has positive terms with consecutive ratio less than s^N when $s < 1$. The tail is bounded by the first omitted term divided by $1 - s^N$. For the endpoint 1, use $I_{N,k}(1) = N^{-1}\mathrm{B}(k/N, 1/2)$ and the preceding half-interval method. No integration across an unbounded endpoint singularity is performed numerically without a remainder bound.

For Gaussian primitives, Taylor's theorem gives

$$\left| J_1(x) - \sum_{j=0}^{m} \frac{(-1)^j x^{2j+1}}{(2j+1)\,j!} \right| \le \frac{x^{2m+3}}{(2m+3)(m+1)!}, \qquad x \ge 0, \tag{14.3}$$

with the sign of the first omitted term. Negative arguments are handled by oddness. Arctangents are reduced to arguments of absolute value less than one and enclosed by alternating series. A rational enclosure of π follows from

$$\pi = 16\arctan(1/5) - 4\arctan(1/239). \tag{14.4}$$

The lemniscate calculation uses the exact coefficient-square sum and Cauchy bound in (7.7)–(7.8).

For a proposed root interval $[u, v]$, the verifier proves that the residual is strictly negative at u and strictly positive at v. The mathematical monotonicity theorem supplies existence and uniqueness inside the interval. The printed midpoint is never used as a proof of its own correctness.

14.2 Verification scripts rerun for the unified edition

The fourteen verification programs contained in the supplied research archives were inspected and rerun with their default exact-arithmetic settings during preparation of this edition. All fourteen completed successfully. A second execution of each program produced a byte-identical JSON result, checked by SHA-256. The optional floating-point mode of the gamma-transfer program was not needed for these reruns.

The following table identifies the checks and records short prefixes of the output hashes. These prefixes are audit identifiers, not mathematical certificates in themselves. The full archived programs contain the rational endpoints; a compact representative verifier is included in Section C so this single-file edition contains a runnable example without requiring external data.

Verification family	Objects checked in the supplied implementation	SHA-256 prefix
Ein area	Exact convolution, eight-equation system, denominator bounds, and a factorial-series enclosure	`083e681137ea`
Gaussian mixtures	Five exact candidate classifications, two root certificates, and reparametrized E-identities	`b5576f7986fa`

Verification family	Objects checked in the supplied implementation	SHA-256 prefix
Student and beta	Four root intervals; 440 half-integral beta parameter pairs; exact differential identities	`6057b3ca1d26`
Shortest beta intervals	2,025 parameter triples; Bernstein signs and coefficient identities; endpoint and length enclosures	`2e676a5f24d1`
Lemniscate length	Order-240 rational series, 964 coefficient equalities, and a closed-form tail majorant	`a50ffd08d46e`
Normalized length	The same coefficient-square representation without numerical division by an approximate π	`2282da708d00`
Prime divisions	Three prime-leaf root enclosures and finite CM-type checks, kept distinct from the general proof	`981cf8464013`
All cyclotomic divisions	118 exponent projectors, 599 polynomial remainders, 7,257 convolution checks, three root intervals	`664439f0f33e`
Algebraic ratios	Two normalized integrals, a golden-ratio inverse, and exact cyclotomic substitutions	`b0f6c3c92576`
Rational inputs	Seven numerical examples, good-prime choices, and 3,500 finite valuation checks	`58fda457e012`
Bounded degree	Three quadratic-input integrals; 18 exact quadratic field calculations and 18,000 valuation inequalities	`50a3aea763a7`
First-kind Fermat beta	Three CDF values and three medians, 21 prime pairs, and 243 polynomial-reduction identities	`1a8b36e134de`
Second-kind beta	Three CDF values and three medians, 27 prime pairs, and first/second-kind character counts	`40eb9e56a1ab`
Gamma transfer	58 polynomial tests, series identities, 484 Laurent coefficient checks, and boundary valuation tests	`58179fa70170`

The equal-output test establishes reproducibility of these implementations, not independent mathematical proof: the same mistaken formula could reproduce the same output. The core inequalities and algebraic identities are therefore proved in the text rather than inferred from these counts.

14.3 Representative certification scales

The supplied verifiers certify the Ein area by an error radius less than 1.76×10^{-210} at order 160. The two Gaussian medians use intervals of width 10^{-66}; the four classical quantile examples use width 10^{-60}. The beta HPD example and the principal arc-division examples use intervals of width at most 10^{-70}. At order 240, the length L_3 has a rational enclosure of width less than 2.28×10^{-83}, and L_3/π one of width less than 8×10^{-84}. The rational-input, bounded-degree, and weighted-beta CDF examples use rational enclosures of width below 10^{-100}. The two algebraic-position ratios from Section 9 are enclosed within widths at most 2×10^{-90}; the golden-ratio inverse uses width 10^{-65}. Displayed decimals may use fewer digits and have wider outward-rounded display intervals than the underlying rational certificates. These two notions of width must not be interchanged.

15 Synthesis and unresolved questions

The most reusable step in this manuscript is not an isolated special-value identity. It is the identification of an arithmetic relation that can be transported between valid settings. Differential translations distinguish an antisymmetric iterated integral from products of its lower-level primitives. Reparametrization puts

Gaussian integrals inside the correct E-function class. A closed square-root lift treats arc length without assuming that sums or limits of transcendental numbers remain transcendental. Cyclotomic period injectivity replaces simplicity where a Jacobian decomposes. Torsion is the algebraic property that can be moved from complex logarithms to local logarithms. Generalized Jacobians retain finite polar information while projecting away additive directions only when the period argument justifies it.

Several boundaries remain explicit. The three possible half-integral beta pairs do not by themselves enumerate every integer parameter pair that realizes the off-center exception. The algebraic-division exceptional set for arbitrary composite exponent is finite, but is not shown here to be empty for every algebraic input. The bounded-degree theorem has a large explicit bound and does not classify all low-degree inputs at small exponents. The polynomial-weight Fermat result does not automatically extend to arbitrary rational weights, equal denominators, or every composite denominator. Nor do the single-value results prove algebraic independence among unrelated constants.

The lemniscate length and rational-division arguments address the precise questions posed in [7, Remarks 8.16 and 5.7], respectively, in the form of proof manuscripts developed here. The fact that a question appears in a 2019 source does not establish that it remained unanswered until this edition. No assertion of first publication or community acceptance is inferred from unsuccessful searches for a matching paper. Independent review should focus on the geometric square-root lift, the exact rational-character projection, the hypotheses of the torsion classifications, and the passage from generalized logarithms to ordinary abelian projections.

For Euler's constant the conclusion is different. The transfer identifies coefficient locking and the arithmetic nature of the missing step, but does not supply that step. The analytic identity $\Psi_{r,0}(1) = e(\gamma - r)$ is exact; a theorem preventing its vanishing for every algebraic r is still needed. Conjecture 13.3 and (13.20) remain conditional targets, not inputs that may be silently assumed. Consequently the manuscript ends with an audited obstruction rather than a claimed proof of irrationality or transcendence of γ.

A Catalogue of explicit constants

The entries below are exact numbers defined by the indicated integral or inverse-function condition. Decimal strings are truncated for orientation. Their arithmetic conclusions refer to the proofs in the cited sections, not to the terminating decimal approximations. Letters used for local examples have their definitions in those sections.

Exact definition	Initial decimal digits	Section
$H_{12}(1)$, antisymmetric Ein area	0.06689807721166676186223 5 . . .	3
Median of the asymmetric truncated Gaussian mixture on $[-1, 1]$	0.11110201921617379287045 2 . . .	4
$J_1(m) = J_1(1)/2$, $0 < m < 1$	0.39269762579294721454724 3 . . .	4
$Q_3(39/40)$ for Student's t law	3.18244630528370959272322 5 . . .	5
Median of Beta$(3/2, 5/2)$	0.35245227987713081376559 9 . . .	5
$1/40$-quantile of Beta$(3/2, 5/2)$	0.03874761778516516078070 1 . . .	5
$39/40$-quantile of the same law	0.82326390286874268256288 1 . . .	5
Left endpoint, 95% shortest interval for Beta$(3/2, 5/2)$	0.00955687087304494286021 9 . . .	6
Right endpoint of the same shortest interval	0.77075726925733542833483 6 . . .	6
Length of the same shortest interval	0.76120039838429048547461 6 . . .	6
Midpoint of the same shortest interval	0.39015707006519018559752 7 . . .	6
$L_3 = \text{length}\{\lvert z^3 - z\rvert = 1\}$	6.46470632781347726562312 6 . . .	7

Exact definition	Initial decimal digits	Section
L_3/π	2.057779935417939294967383 . . .	7
$s_{8,1/2}$, four-leaf half-arc midpoint	0.581372564921331418342397 . . .	8
$s_{10,1/2}$, five-leaf half-arc midpoint	0.566067311893637997084837 . . .	8
$s_{12,1/2}$, six-leaf half-arc midpoint	0.555521188888391003316451 . . .	8
$s_{18,1/2}$, nine-leaf half-arc midpoint	0.537498666292662410093407 . . .	8
$R_8(1/2)$	0.429797068135111779727193 . . .	9
$R_{10}(1/2)$	0.441595303662667131493307 . . .	9
$s_{8,(\sqrt{5}-1)/2}$	0.716302719414143509845268 . . .	9
$R_{12}(1/2)$	0.450017109099808036111667 . . .	10
$R_{18}(1/2)$	0.465117308242266039709804 . . .	10
$R_{24}(1/2)$	0.473209591762729788013422 . . .	10
$R_9(1/2)$	0.436230919677225329059895 . . .	10
$R_{12}(2^{-1/4}) = \mathrm{I}_{1/8}(1/12, 1/2)$	0.760658039134571678695270 . . .	10
$R_{12}(3-\sqrt{7})$	0.318832995555007621109640 . . .	11
$R_{18}(3-\sqrt{7})$	0.329534360277977105299820 . . .	11
$R_{24}(3-\sqrt{7})$	0.335267754547646648453111 . . .	11
$\mathrm{I}_{1/2}(1/3, 1/5)$	0.367139793949012364863668 . . .	12
Median of Beta$(1/3, 1/5)$	0.804376357494993257372466 . . .	12
$\mathrm{I}_{1/2}(4/3, 6/5)$	0.460657366225158194044108 . . .	12
Median of Beta$(4/3, 6/5)$	0.534154161604607302042768 . . .	12
CDF at $1/2$ for $a = 1/3, b = 1/5, W = 1 + t^2$	0.248596392472207792663111 . . .	12
Median for $a = 1/3, b = 1/5, W = 1 + t^2$	0.938730288889941676673362 . . .	12
$\mathrm{I}_{1/2}(2/3, 2/5)$	0.35389888866285687131 . . .	12
Median of Beta$(2/3, 2/5)$	0.71685421441072752258 . . .	12
$\mathrm{I}_{1/2}(5/3, 7/5)$	0.430643543621647954 97 . . .	12
Median of Beta$(5/3, 7/5)$	0.553989284028517005618288 . . .	12
CDF at $1/2$ for $a = 2/3, b = 2/5, W = 1 + t^2$	0.25205276478993036618 . . .	12
Median for $a = 2/3, b = 2/5, W = 1 + t^2$	0.83890381650993691841 . . .	12

No row in this catalogue is a claim about algebraic independence between separate families. In particular the gamma constant itself is deliberately not entered as a newly classified number.

B Concordance with the research notes

This single paper consolidates fourteen supplied notes. The table gives English descriptive titles and their destination in the unified text. These notes are internal research manuscripts, not external peer-reviewed references; external inputs are listed separately in the bibliography. Repetition has been removed by incorporating the strongest applicable version while retaining earlier examples and explicit failures.

Note	Subject of the supplied source	Unified location
N01	Antisymmetric Ein iterated integrals and algebraic independence	Section 3
N02	Truncated Gaussian mixtures and a finite algebraic-quantile test	Section 4
N03	Student, half-integral beta, Jeffreys, and Clopper–Pearson quantiles	Section 5
N04	Arithmetic dichotomy for shortest beta probability intervals	Section 6
N05	Polynomial lemniscate lengths as closed periods	Section 7

Note	Subject of the supplied source	Unified location
N06	Prime-leaf rational divisions by simple-Jacobian rigidity	Subsumed by Section 8
N07	Normalized lemniscate length and logarithm separation	Section 7
N08	Cyclotomic proof for all integral leaf counts and exponents	Section 8
N09	Algebraic division ratios, cyclotomic conjugates, and torsion	Section 9
N10	Rational radii, Zsigmondy primes, and local nonvanishing	Section 10
N11	Height-independent bounds for algebraic inputs of bounded degree	Section 11
N12	First-kind Fermat beta integrals and polynomial-weight stability	Section 12
N13	Second-kind beta forms and generalized-Jacobian projection	Section 12
N14	Transfer audit for Euler's constant and the missing mixed-arithmetic step	Section 13

The N06 and N07 notes shared one computational archive, while the associated total-length and normalized-length programs remain separate. The gamma audit also recalled a September exploratory record on ten approaches. That record is contextual rather than an additional source of proved transcendence results. Its principal lessons—that finite experiments do not prove impossibility, that denominator savings must beat total arithmetic cost, and that nonsplitting alone does not determine a comparison-matrix entry—are retained in Section 13.

C A compact exact-arithmetic verifier

The following self-contained Python listing uses only the standard library. It verifies selected coefficient and valuation identities, then certifies $R_{12}(1/2)$ in a rational interval of width below 10^{-100}. The printed 60-place enclosure is intentionally wider than the internal interval. It was executed in this form for this edition. It does not implement the analytic subgroup theorem and is not a formal proof of transcendence.

```
"""Exact rational checks; Python 3.10+, standard library only.
This code certifies a numerical interval, not transcendence.
"""
from fractions import Fraction as Q
from math import comb, factorial


def iroot(m: int, n: int) -> int:
    if m < 0 or n < 1:
        raise ValueError("Nonnegative radicand and positive degree required")
    lo, hi = 0, 1 << ((m.bit_length() + n - 1) // n)
    while lo + 1 < hi:
        mid = (lo + hi) // 2
        if mid ** n <= m:
            lo = mid
        else:
            hi = mid
    return hi if hi ** n <= m else lo


def root_bounds(q: Q, n: int, digits: int = 110):
    if q < 0:
        raise ValueError("Positive real root expected")
    scale = 10 ** digits
    m = iroot(q.numerator * scale ** n // q.denominator, n)
```

```
    return Q(m, scale), Q(m + 1, scale)


def beta_half(a: Q, b: Q, terms: int = 350):
    if not (0 < a < 1 and 0 < b < 1):
        raise ValueError("This positive-tail bound assumes 0 < a,b < 1")
    term, total = 1 / a, Q(0)
    for j in range(terms):
        total += term
        term *= (1 - b + j) * (a + j)
        term /= 2 * (j + 1) * (a + j + 1)
    lo, hi = root_bounds(Q(1, 2 ** a.numerator), a.denominator)
    # Consecutive terms have ratio < 1/2; term is first omitted.
    return lo * total, hi * (total + 2 * term)


def arc_bounds(n: int, s: Q, terms: int = 80):
    if n < 3 or not 0 < s < 1:
        raise ValueError("Need n >= 3 and 0 < s < 1")
    term, total, power = s, Q(0), s ** n
    for j in range(terms):
        total += term
        term *= Q(2 * j + 1, 2 * j + 2) * power
        term *= Q(n * j + 1, n * (j + 1) + 1)
    return total, total + term / (1 - power)


def ratio_bounds(n: int, s: Q):
    a, b = Q(1, n), Q(1, 2)
    p, q = beta_half(a, b), beta_half(b, a)
    den = ((p[0] + q[0]) / n, (p[1] + q[1]) / n)
    num = arc_bounds(n, s)
    assert den[0] > 0
    return num[0] / den[1], num[1] / den[0]


def decimal_floor(q: Q, digits: int = 60) -> str:
    scale = 10 ** digits
    m = q.numerator * scale // q.denominator
    return str(m // scale) + "." + str(m % scale).zfill(digits)


def check_coefficients():
    # Two independent expressions for coefficients of H_12(z).
    for n in range(2, 41):
        a = Q((-1) ** n, n) * sum(
            (Q(comb(n, k) * (2 ** k - 2 ** (n-k)), k)
             for k in range(1, n)), Q(0))
        ordinary = sum(
            (Q((-1) ** n * (2 ** k - 2 ** (n-k)),
               n * k * factorial(k) * factorial(n-k))
             for k in range(1, n)), Q(0))
        assert a == factorial(n) * ordinary
    # Universal local denominator bound on a finite test window.
    for ell in (5, 7, 11, 13):
        for j in range(1, 501):
            m, v = 2 * j + 1, 0
            while m % ell == 0:
                m //= ell
                v += 1
```

```
            assert (ell - 1) * v <= 2 * j


if __name__ == "__main__":
    check_coefficients()
    lo, hi = ratio_bounds(12, Q(1, 2))
    assert hi - lo < Q(1, 10 ** 100)
    scale = 10 ** 60
    out_lo = Q(lo.numerator * scale // lo.denominator, scale)
    out_hi = Q(-((-hi.numerator * scale) // hi.denominator), scale)
    assert out_lo <= lo < hi <= out_hi
    print("Lower:", decimal_floor(out_lo))
    print("Upper:", decimal_floor(out_hi))
    print("Exact interval width < 10^(-100).")
```

The output is

```
Lower:  0.450017109099808036111667642573989081152617667160799532175248
Upper:  0.450017109099808036111667642573989081152617667160799532175249
Exact interval width < 10^-100.
```

The internal width assertion and the wider decimal output concern different intervals. The rational upper endpoint is rounded upward, and the lower endpoint downward. The verification of finitely many local inequalities is supplemental to the proof valid for every index in Lemma 10.1.

References

[1] F. Beukers, *A refined version of the Siegel–Shidlovskii theorem*, Annals of Mathematics **163** (2006), 369–379. doi:10.4007/annals.2006.163.369. Theorems 1.1 and 1.3 and Corollary 1.4.

[2] V. Ravi Srinivasan, *Extensions by Antiderivatives, Exponentials of Integrals and by Iterated Logarithms*, preprint, 2008. `https://arxiv.org/abs/0811.3004`. The Kolchin–Ostrowski theorem in Section 2 is the form used here.

[3] A. Baker, *Linear forms in the logarithms of algebraic numbers (III)*, Mathematika **14** (1967), 220–228. doi:10.1112/S0025579300003843.

[4] A. Huber and G. Wüstholz, *Transcendence and Linear Relations of 1-Periods*, Cambridge Tracts in Mathematics 227, Cambridge University Press, 2022. Author manuscript: `https://arxiv.org/abs/1805.10104`. Theorem 6.2, the generalized-Jacobian framework, and Corollary 13.13 are used with their stated hypotheses.

[5] J. S. Milne, *Algebraic Groups: The Theory of Group Schemes of Finite Type over a Field*, Cambridge University Press, 2017. In particular the Barsotti–Chevalley structure theorem. `https://www.jmilne.org/math/Books/AGbook.pdf`.

[6] S. Fischler and T. Rivoal, *Values of E-functions are not Liouville numbers*, Journal de l'École polytechnique—Mathématiques **11** (2024), 1–18. doi:10.5802/jep.249. `https://arxiv.org/abs/2301.01158`. Theorem A restates the quantitative result used for the Ein example.

[7] M. Tamiozzo, *Algebraicity of the division points of the trifolium and related topics*, preprint, 2019. `https://arxiv.org/abs/1912.00204`. The arc-length formula in Section 3.4, the small-leaf results, and Remarks 5.7 and 8.16 provide the historical problem statements discussed here.

[8] S. Yu. Orevkov, *Irreducibility of lemniscates*, preprint, 2018. `https://arxiv.org/abs/1806.10377`. The needed irreducibility argument is also given directly in the text.

[9] A. Carocca and R. E. Rodríguez, *Jacobians with group actions and rational idempotents*, Journal of Algebra **306** (2006), 322–343. `https://arxiv.org/abs/math/0305328`.

[10] J. Boxall and D. Grant, *Examples of torsion points on genus two curves*, Transactions of the American Mathematical Society **352** (2000), 4533–4555. doi:10.1090/S0002-9947-00-02368-0. Theorem (0.3), the 22-point packet for $y^2 = x^5 + x$.

[11] V. Arul, *Torsion points on Fermat quotients of the form $y^n = x^d + 1$*, preprint, 2019, revised version 2020. `https://arxiv.org/abs/1910.14251`. Definition 1.0.7, Proposition 2.4.4, Corollary 2.4.5, and Theorem 6.2.21. The geometric torsion classification, not just rational-point torsion, is the input required here.

[12] D. Roessler, *A note on the Manin–Mumford conjecture*, preprint, 2004. `https://arxiv.org/abs/math/0409083`. Used for the torsion-coset formulation, not for an effective bound.

[13] Yu. G. Zarhin, *p-adic abelian integrals and commutative Lie groups*, preprint, 1996. `https://arxiv.org/abs/alg-geom/9603006`. Sections 1 and 3: functorial logarithms and local abelian integrals.

[14] R. F. Coleman, *Torsion points on curves and p-adic Abelian integrals*, Annals of Mathematics **121** (1985), 111–168. doi:10.2307/1971194. Background for the local torsion strategy; the local series in this paper is derived explicitly.

[15] Ö. Avcı, *Large Zsigmondy Primes*, preprint, 2020, revised version 2023. https://arxiv.org/abs/2011.06136. Only the classical Bang–Zsigmondy theorem recalled in the introduction is needed here, not the paper's stronger large-prime result.

[16] K. Zsigmondy, *Zur Theorie der Potenzreste*, Monatshefte für Mathematik und Physik **3** (1892), 265–284. doi:10.1007/BF01692444.

[17] P. M. Voutier, *Primitive divisors of Lucas and Lehmer sequences, II*, Journal de Théorie des Nombres de Bordeaux **8** (1996), 251–274. doi:10.5802/jtnb.168. Author version: https://arxiv.org/abs/1211.3107. Theorem 2(i) is used; its ratio-degree parameter is distinguished from the degree of an auxiliary ambient field.

[18] U. Zannier, *Hyperelliptic Continued Fractions and Generalized Jacobians*, preprint, 2016. https://arxiv.org/abs/1602.00934. Section 2.1.2 describes the modulus extension used in the second-kind argument.

[19] D. Agostini, T. Ö. Çelik, and J. B. Little, *Polynomial Analogs of Theta Functions and Rational Solutions of the KP Equation*, International Mathematics Research Notices **2024** (22), 13964–13987. doi:10.1093/imrn/rnae227. Sections 2.1 and 2.3 supply the Rosenlicht and Abel-map descriptions.

[20] S. Fischler and T. Rivoal, *Relations between values of arithmetic Gevrey series, and applications to values of the Gamma function*, Journal of Number Theory **261** (2024), 36–54. doi:10.1016/j.jnt.2024.02.016. https://arxiv.org/abs/2301.13518. Conjectures 1 and 3, Definition 1, and Theorem 2 are explicitly separated in the gamma audit.

[21] T. Rivoal, *Les E-fonctions et G-fonctions de Siegel*, Journées X-UPS 2019. doi:10.5802/xups.2019-03. Theorem 5.11 and its discussion of e, Ein(1), and the Euler–Gompertz alternative.

[22] S. Dhillon and E. Saha, *On arithmetic nature of special values of the incomplete beta function*, Acta Arithmetica **218** (2025), 273–284. doi:10.4064/aa240610-6-10. Related literature; a complete theorem-by-theorem priority comparison with this paper has not been performed.

[23] NIST Digital Library of Mathematical Functions, Chapters 5, 6, 7, and 8: gamma, beta, exponential-integral, error-function, and incomplete-beta identities. https://dlmf.nist.gov/. Used for standard definitions and functional identities, not as a source for the new applications assembled in this manuscript.

[24] NIST/SEMATECH, *e-Handbook of Statistical Methods*, Sections 1.3.6.6.5, 1.3.6.6.17, and 1.3.6.7.2: F distribution, beta distribution, and Student's t critical values. https://www.itl.nist.gov/div898/handbook/.

[25] P. Laud, *ratesci: Confidence Intervals for Comparisons of Binomial or Poisson Rates*, CRAN package documentation, vignette "1: Single rate" and reference page jeffreysci. https://cran.r-project.org/web/packages/ratesci/. The distinction between raw posterior quantiles and boundary-adjusted interval endpoints is retained in the text.

[26] N. Lourie, K. Cho, and H. He, *Show Your Work with Confidence: Confidence Bands for Tuning Curves*, preprint, 2023. https://arxiv.org/abs/2311.09480. Appendix C: the beta highest-density interval and equal-endpoint-density condition.